Strong Predictors, Weak Selectors: Why Confidence Signals Fail to Improve LLM-Based Named Entity Recognition

Anonymous ACL submission

Abstract

Self-consistency yields large gains on discrete-answer tasks, but named entity recognition (NER) produces *sets* of entities with partial overlap, where no majority vote analog exists. We evaluate five confidence signals for quality estimation (QE) and sample selection on four NER benchmarks using 4B–8B models. All signals achieve reliable QE (ρ up to 0.45, AUROC up to 0.82), enabling selective prediction ($+5$–8 pp F1 at 80% coverage). For instance-level selection, LP achieves $+1.39$ pp on Few-NERD ($p<0.001$) where LP-quality alignment is strong, but gains are absent elsewhere despite $\sim$13 pp oracle headroom (the **correlation-selection gap**). We diagnose two causes: instance degeneracy (13–87%) and weak LP-quality alignment; degeneracy causally modulates LP within fixed settings but does not predict cross-dataset direction (ρ=0.052, 23 configurations). The correct self-consistency analogy for structured outputs is entity-level *construction*: aggregating entities across samples improves F1 by $+0.88$ to $+1.67$ pp ($p<0.005$) on datasets where instance-level selection fails or is marginal. Construction and LP selection are complementary, governed by LP-quality alignment; an alignment-based gating protocol correctly routes all three evaluated datasets.

1 Introduction

Self-consistency (majority voting over multiple samples) yields large gains on discrete-answer reasoning (Wang et al., 2023), and test-time compute scaling has emerged as a complementary axis to model scaling (Snell et al., 2025; Brown et al., 2024). But named entity recognition (NER) poses a structurally different challenge: the output is a *set* of entities, two samples may extract overlapping but non-identical sets, and there is no obvious majority to vote on.

Prior work on confidence estimation for LLM-based IE has conflated *quality estimation* (which inputs does the model handle reliably?) with *sample selection* (which of N outputs is best?). Existing self-consistency methods (Wang et al., 2023; Chen et al., 2024; Taubenfeld et al., 2025; Kang et al., 2025) assume outputs can be grouped by equivalence, an assumption that breaks for entity sets with partial overlap. Unlike Yao et al. (2026), who demonstrate that self-consistency correlates with task accuracy, we provide the first systematic empirical characterization of *when and why* LP-based selection succeeds or fails for structured prediction, diagnosing degeneracy and compression as the two blocking mechanisms. Our experiments use fine-tuned models, the dominant deployment paradigm for production NER, and separate quality estimation from best-of-N selection.

A key distinction structures our analysis: four of five signals (SJ, VC, FK, EM) are instance-level aggregates that assign identical scores to all N candidates for a given input; only log-probability (LP) is natively per-sample. All signals achieve reliable quality estimation (ρ up to 0.45, AUROC up to 0.82; Figure 1), supporting selective prediction. For selection, LP achieves $+1.39$ pp on Few-NERD ($p<0.001$) in settings where both low instance degeneracy and adequate alignment are present, with gains scaling with N. On SciERC, the observed LP gain ($+0.2$ pp, p=0.29) falls below the minimum detectable effect ($\approx$1.5 pp at 80% power, n=529) despite 13 pp oracle headroom. We term this disparity between quality estimation strength and selection effectiveness the **correlation-selection gap**. Two factors shape this gap: (a) *instance degeneracy*, the fraction of instances where all N samples produce identical F1, (13–87% across our benchmarks) limits the available headroom; (b) *LP-quality alignment* (within-instance $\rho(\text{LP}, \text{F1})$) determines whether LP can exploit the remaining headroom (§5). Our diagnosis is complementary to MBR utility misalignment (Eikema et al., 2025): where MBR failures in structured prediction stem from poorly chosen util-

ity functions, we find that degeneracy eliminates candidate diversity before utility alignment matters. Within non-degenerate instances, MBR utility correlates positively with F1 (ρ=0.52; §5). Both mechanisms intensify with model capability: degeneracy rises with scale and training convergence (§5). This creates a tension between model quality and test-time improvability.

We evaluate five signals from three families (structural, surface, and model-internal) on SciERC (Luan et al., 2018) as a primary medium-difficulty test bed, with Few-NERD (Ding et al., 2021) (low-degeneracy contrast), CoNLL-2003 (Sang and Meulder, 2003) (ceiling-effect control), and WNUT-17 (Derczynski et al., 2017) (signal-collapse boundary), using two 8B models (Qwen3-8B, LLaMA-3.1-8B) with a 4B ablation (Qwen3-4B). We vary candidate pool size ($N \in \{2, \ldots, 32\}$), signal sophistication, and model capacity within the 4B–8B range; behavior at 70B+ scales remains an open question. An exploratory extension to relation extraction is reported in Appendix L. Our contributions:

1. **Quality estimation framework and selective prediction.** Five signals from three families evaluated on four NER benchmarks, two model families, and a 4B ablation, achieving reliable quality estimation (ρ up to 0.45, AUROC up to 0.82). QE enables selective prediction: filtering low-confidence instances at 80% coverage yields +5–8 pp F1 (AUPRC 0.70–0.92, VC strongest; §4.7).

2. **Diagnostic of the correlation-selection gap.** LP-quality alignment (within-instance $\rho(\text{LP}, \text{F1})$) is the primary predictor of selection success: LP achieves +1.39 pp on Few-NERD (p<0.001) where alignment is strong, but not on benchmarks with weak alignment. Instance degeneracy causally modulates LP within fixed settings (epoch and temperature ablations), but does not predict cross-dataset direction (ρ=0.052, 23 configurations); structural signals fail by construction; supervised combinations and hidden state probes (§4.6) confirm the gap persists.

3. **Entity-level construction as the correct self-consistency analogy.** Rather than selecting one of N outputs, LP-weighted entity aggregation yields +1.67 pp on SciERC (p=0.004) and +0.88 pp on Few-NERD (p<0.001; §4.7). Construction and LP selection are complementary, governed by LP-quality alignment; a simple alignment-based gating protocol correctly routes all three evaluated datasets.

2 Related Work

Self-consistency and test-time scaling. Self-consistency (Wang et al., 2023) improves reasoning via majority voting; extensions handle free-form text (Chen et al., 2024; Thirukovalluru et al., 2024; Manakul et al., 2023). Test-time scaling studies (Snell et al., 2025; Brown et al., 2024) and reward-free selectors (Taubenfeld et al., 2025; Kang et al., 2025; Wang et al., 2025a; Oh and Lee, 2026) rely on answer-level agreement. Adaptive variants gate on answer agreement (Aggarwal et al., 2023), input difficulty (Wang et al., 2025b; Feng et al., 2025), or response confidence (Kim et al., 2026); Chen et al. (2025) diagnose SC failure modes without proposing gating. All target discrete-answer reasoning; structured prediction requires different gating signals because degeneracy operates at the entity-set level. Li et al. (2025) propose generative self-aggregation for IE but focus on fusion rather than selection.

Consistency for structured IE. Yao et al. (2026) show consistency correlates with accuracy for LLM-based NER/RE; we show correlation alone does not ensure selection gains, diagnosing degeneracy and compression as blocking mechanisms. Our experiments use fine-tuned models, where degeneracy is most pronounced and is modulated by task structure and training convergence (Appendix Z). CoCoA (Vashurin et al., 2025) fuses confidence with consistency for QA/summarization; our supervised signal combination baseline (§4.5) suggests combination strategy is not the bottleneck. Earlier work on confidence for structured prediction (Culotta and McCallum, 2004; Mejer and Crammer, 2010; Fomicheva et al., 2020) establishes the paradigm we extend to set-valued outputs. Our evaluation builds on generative IE (Zhou et al., 2024; Sainz et al., 2024; Li et al., 2023; Wadhwa et al., 2023) and constrained decoding (Willard and Louf, 2023; Dong et al., 2024; Park et al., 2024; Schall and de Melo, 2025). Yun et al. (2025) show structured formats induce diversity collapse; our LP compression (median range 0.041 nats; §5) corroborates this for NER.

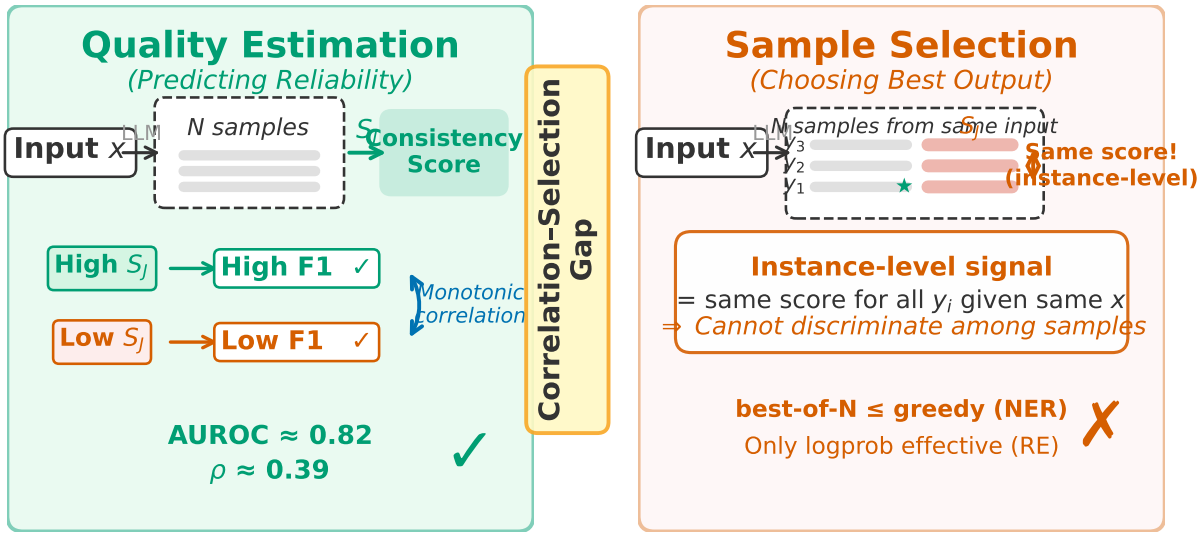


Figure 1: The correlation-selection gap. **Left**: consistency signals reliably predict instance-level quality (ρ up to 0.45 on CoNLL, 0.38 on the primary test bed, 0.38 on Few-NERD; AUROC up to 0.82, CoNLL EM). **Right**: quality estimation does not translate to selection; all methods approximately match greedy decoding (T=0) despite 13 pp oracle headroom (§5).

Uncertainty estimation and MBR decoding. Surveys cover confidence estimation (Geng et al., 2024), uncertainty quantification (Xia et al., 2025), and calibration (Hobelsberger et al., 2025); semantic uncertainty (Kuhn et al., 2023; Farquhar et al., 2024) and P(True) (Kadavath et al., 2022) provide model-internal alternatives. Our selection strategies instantiate MBR decoding (Eikema and Aziz, 2022; Deguchi et al., 2024); unlike MT where MBR yields consistent gains (Freitag et al., 2022) and code generation where execution-based MBR improves selection (Shi et al., 2022), instance degeneracy eliminates the headroom MBR needs even when utility is well-aligned (ρ=0.52; §5). Eikema et al. (2025) find MBR utilities degrade on structured outputs; our diagnosis is complementary: degeneracy, not utility misalignment, is the primary bottleneck. Our QE results connect to selective prediction (Geifman and El-Yaniv, 2017; Kuleshov and Liang, 2015; Varshney et al., 2022): signals with AUROC up to 0.82 support abstention policies independent of sample selection.

Best-of-N with learned verifiers. Gao et al. (2023) establish BoN scaling laws and Chen et al. (2021) define the pass@k metric for code generation; our unsupervised signals face a more fundamental limitation than reward overoptimization. Trained verifiers (Cobbe et al., 2021; Lightman et al., 2024; Fernandes et al., 2022) learn per-sample scoring; reward model ensembles mitigate but do not eliminate reward hacking (Eisenstein et al., 2023). Our LoRA-rank ablation (degeneracy 7.2%, no selection improvement at ρ≈0.24) suggests increasing diversity alone is insufficient within unsupervised signals. Fine-tuning contributes to degeneracy (Kirk et al., 2024), and constrained structured outputs limit diversity recoverable by temperature scaling (Holtzman et al., 2020) or diverse decoding (Vijayakumar et al., 2018).

3 Method

3.1 Problem Formulation

Given model $\mathcal{M}$ and input x, we sample N outputs $\{y_1, \ldots, y_N\}$ at temperature $T > 0$ via constrained decoding (Willard and Louf, 2023), where each y_i is an entity set. A **per-sample signal** $c_{\text{ps}} : (x, y_i) \to \mathbb{R}$ scores each output independently; an **instance-level signal** $c_{\text{il}} : (x, \{y_i\}_{i=1}^{N}) \to \mathbb{R}$ produces a single score per input (notation in Appendix A). This distinction is consequential for selection: in our study, 4 of 5 signals are instance-level aggregates that assign identical QE scores to all N candidates. Consensus-proximity (MBR) converts these to per-sample rankings (§4.3), but these rankings do not improve selection; only natively per-sample signals can, in principle, exploit oracle headroom.

3.2 Confidence Signals

Four are *consistency signals*: *structural* (Soft Jaccard [SJ], entity voting confidence [VC]) and *surface* (exact match [EM], Fleiss' κ [FK]); log-probability [LP] is the sole *model-internal*, per-sample signal.

Log-probability (model-internal, per-sample). For sample y_i with tokens $t_1, \ldots, t_{|y_i|}$:

$$c_{\text{lp}}(y_i) = \frac{1}{|y_i|} \sum_{j=1}^{|y_i|} \log p_{\mathcal{M}}(t_j \mid t_{<j}, x) \quad (1)$$

Exact match voting (surface, instance-level). The proportion of samples producing the most common output:

$$c_{\text{em}}(x) = \frac{1}{N} \max_y \left|\{i : y_i = y\}\right| \quad (2)$$

Fleiss' κ (surface, instance-level). We adapt Fleiss (1971) to entity sets: items are unique (text, type) tuples across all N samples; each sample assigns binary present/absent labels, yielding a $|E_\cup| \times 2$ rating matrix:

$$\kappa = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e} \quad (3)$$

where $\bar{P}$ is mean observed and $\bar{P}_e$ expected agreement. Temperature-based sampling introduces mild negative dependence (-0.4% to -4.0%; Appendix B), which does not inflate FK. Unlike structural signals, FK relies on exact string matching.

Soft Jaccard (structural, instance-level). For samples y_i, y_j, we group extractions by type, compute max-weight bipartite matching (Hungarian algorithm, token-level IoU), and define:

$$\text{SJ}(y_i, y_j) = \frac{\sum_{m \in M} \text{sim}(a_m, b_m)}{\sum_g \max(|g_i|, |g_j|)} \quad (4)$$

where M is the matched set, $\text{sim}(a_m, b_m)$ is the token-level IoU, and $\max(|g_i|, |g_j|)$ penalizes length asymmetry.[1] The instance-level QE signal averages over all pairs: $c_{\text{sj}}(x) = \binom{N}{2}^{-1} \sum_{i<j} \text{SJ}(y_i, y_j)$. For selection, consensus-proximity converts this to per-sample utility $u_{\text{sj}}(y_i) = (N-1)^{-1} \sum_{j \neq i} \text{SJ}(y_i, y_j)$, equivalent to MBR decoding (Eikema and Aziz, 2022). The same construction applies to FK, EM, and VC.

[1]The two normalizer variants produce nearly identical rankings ($|\Delta\rho| \leq 0.009$; Table 10).

Entity voting confidence (structural, instance-level). For each entity e in the union $E_\cup$ of all samples, we compute frequency $\text{freq}(e) = |\{i : e \in y_i\}|/N$. The instance-level confidence is the mean frequency:

$$c_{\text{vote}}(x) = \frac{1}{|E_\cup|} \sum_{e \in E_\cup} \text{freq}(e) \quad (5)$$

When $E_\cup = \emptyset$, $c_{\text{vc}}(x) = 0$.[2]

3.3 Evaluation Framework

Quality estimation. Spearman ρ between the signal and instance-level F1, and AUROC classifying instances as above/below median F1. Both quality estimation and selection exclude instances with no gold annotations: n=529 on SciERC (from 551), n=2,756 on CoNLL (from 3,453).[3]

Sample selection. Best-of-N F1: per-sample signals (LP) directly rank candidates; instance-level signals rank via consensus-proximity (MBR utility; see SJ definition above). Baselines: *oracle* (highest-F1 sample) and *greedy* (T=0 argmax decoding). We define **instance degeneracy** as the fraction of instances where all N samples yield identical F1; for these instances, no signal can improve selection. We report F1 under the protocol matching each experiment's configuration (seed, epoch, N). SciERC greedy F1 appears as four values across tables (0.640, 0.644, 0.645, 0.650) due to protocol variants: primary seed vs. multi-seed mean, 5-epoch vs. 3-epoch checkpoint, and gold-filtered vs. full test set. Each table specifies its protocol.

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate on four English NER benchmarks spanning a difficulty gradient, each serving a distinct role. **SciERC** (Luan et al., 2018) (551 test instances, 6 entity types) is the primary medium-difficulty test bed where models achieve ≈65% F1. **Few-NERD** (Ding et al., 2021) (INTRA coarse; 37,648 instances, 8 entity types, ≈75% F1) provides a low-degeneracy contrast where LP selection succeeds (§4.4). **CoNLL-2003** (Sang and Meulder, 2003) (3,453 test sentences, 4 entity

[2]For voted-output *selection*, we include only entities with frequency $> N/2$ (standard majority voting).

[3]On CoNLL, full-set (n=3,453) Spearman ρ values are higher than gold-filtered (n=2,756) by +0.03–0.05 due to the zero/non-zero boundary inflating correlations, paralleling the effect documented for WNUT-17 in Appendix D.

types, ≈91% F1) serves as a ceiling-effect control where limited oracle headroom constrains selection gains. **WNUT-17** (Derczynski et al., 2017) (1,287 social media instances, 52% zero F1) probes the signal-collapse boundary where model capability is insufficient.

Models and sampling. Both **Qwen3-8B** and **LLaMA-3.1-8B-Instruct** are fine-tuned with LoRA (rank=32, α=64) for 5 epochs using LLaMA Factory (Zheng et al., 2024) under identical training procedures, except Few-NERD models which use 3 epochs (early stopping at the best validation checkpoint; a 5-epoch control confirms LP selection remains positive though gains are halved; Appendix Z). Qwen3-8B's SciERC checkpoint has a minor train–test mismatch (empty thinking tokens); a quantitative analysis rules out this confound (Appendix I). We draw N=8 samples at T=1.0 (primary), with N=16 for scaling experiments (ranking stability $\sigma \leq 0.026$, Table 17; sublinear gains beyond N=8, Appendix R). Inference uses vLLM with constrained decoding (Willard and Louf, 2023) (seed protocol: Appendix G).

4.2 Quality Estimation Results

Table 1 presents ρ and AUROC for five signals across tasks and models (ε-free; ε=0.05 applies to selection only, stable across $[0, 0.2]$, Appendix J).

Signal rankings are difficulty-dependent. On SciERC (medium difficulty), structural signals lead (VC ρ=0.379, SJ 0.360) over LP (0.205); bootstrap CIs confirm the gap ($\Delta\rho$=+0.155; Appendix E). On CoNLL (single-sample F1≈0.91), EM leads both ρ and AUROC (0.816). At the signal-collapse boundary (WNUT-17, macro F1≈42%), full-set correlations are dominated by the zero/non-zero boundary (VC ρ=0.724 but conditional ρ=0.038; Table 11); LP shows the reverse (conditional ρ=0.350; Figure 6).

4.3 The Correlation-Selection Gap

Two factors constrain selection: instance degeneracy collapses the candidate pool for consensus-based approaches, and weak LP-quality alignment prevents exploiting the remaining headroom. Four instance-level signals cannot rank candidates by construction; LP selection effectiveness is primarily determined by LP-quality alignment rather than degeneracy level (§5).

Despite moderate quality estimation, no signal consistently improves selection over greedy (Table 2; Figure 2). Instance-level signals assign identical QE scores to all candidates; MBR conversion does not improve over greedy on the primary test bed.

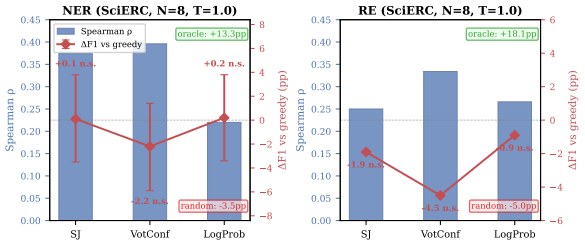


Figure 2: The correlation-selection gap on SciERC NER (N=8, T=1.0, 2-seed mean). Bars show best-of-N selection F1; the line marks quality estimation ρ. Despite ρ up to 0.40, no signal significantly outperforms greedy (all p>0.05; Table 2). Dashed line: oracle upper bound.

No significant improvement over greedy. On SciERC (n=529, 2-seed sign tests[4]), the oracle (0.775) reveals 13.0 pp of headroom, but LP achieves +0.2 pp (p=0.29), below the MDE (≈1.5 pp). The pattern holds across CoNLL and WNUT-17 (Table 12); under stringent TOST anchors, some configurations show significant *degradation* (Appendix F). Few-NERD provides the definitive contrast: LP achieves +1.39 pp (p<0.001) at low degeneracy (§4.4), while structural signals still degrade selection. Instance-level signals predict *which inputs* the model handles reliably, not *which output* is best; closing this gap likely requires trained verifiers (Cobbe et al., 2021) (the sole positive finding, non-tied LP subset +2.9 pp p=0.004, survives Bonferroni correction; §5).

4.4 Cross-Dataset Contrast: Few-NERD

Few-NERD (Table 3) confirms that LP selection effectiveness is inversely associated with degeneracy. Under standard early stopping (3 epochs for Few-NERD vs. 5 for SciERC/CoNLL), Few-NERD degeneracy is 13.8%, and LP achieves +1.39 pp (p<0.001, seed σ=0.04 pp vs. SJ σ=2.02 pp), while all structural signals degrade selection (VC −1.79, SJ −2.14 pp). LP selection is 50× more stable across random seeds than structural selection, making it the only practically reliable signal for deployment scenarios requiring reproducibility. LLaMA-3.1-8B confirms the pattern: LP is the only positive signal (+0.36 pp) despite higher

[4]MDE≈1.5 pp at 80% power (n=529); Few-NERD (n=32,565; MDE≈0.22 pp) provides the definitive test.

Table 1: NER quality estimation (ρ and AUROC; N=8, primary seed=42; 3-seed stability $\sigma \leq 0.026$ in Table 17). Best in **bold**. RE results in Appendix L. N=16 in Table 8.

		SciERC		CoNLL (Qwen)		CoNLL (LLaMA)	
Signal	Family	ρ	AUC	ρ	AUC	ρ	AUC
VC	Structural	**.379**	**.663**	.424	.789	.426	.709
SJ	Structural	.360	.650	.436	.801	**.431**	.704
FK	Surface	.267	.602	.400	.774	.428	.703
EM	Surface	.295	.614	**.451**	**.816**	**.431**	.705
LP	Model	.205	.578	.316	.753	.307	**.761**

Table 2: Best-of-N NER selection (macro F1) on SciERC (N=8, T=1.0, 2-seed mean$\pm\sigma$). LP is per-sample; others use consensus-proximity. Best non-oracle in **bold**. Non-tied subset in §5; RE in Appendix L.

Method	NER F1	p
Oracle	0.775 ± 0.001	—
Greedy (T=0)	0.645 ± 0.001	—
Per-sample signal		
LP	$\mathbf{0.647} \pm 0.005$	.29
Instance-level signals (consensus-proximity selection)		
SJ	0.646 ± 0.002	.67
FK	0.646 ± 0.001	.79
EM	0.637 ± 0.001	—
VC	0.623 ± 0.004	—
Random	0.610 ± 0.003	—

Table 3: LP selection across datasets (Qwen3-8B, N=8, T=1.0, gold-filtered). Instance degeneracy (Degen.%): fraction where all N samples yield identical F1. p: paired bootstrap (n=10,000) vs. greedy. Few-NERD and CoNLL Δ values are 3-seed means; SciERC is single-seed (3-seed mean: $+0.04 \pm 0.48$ pp). Oracle headroom: 5.0 pp (CoNLL), 13.0 pp (SciERC), 12.4 pp (Few-NERD). Minor greedy F1 variations across tables reflect protocol differences (§3.3). Few-NERD uses 3-epoch checkpoints; SciERC and CoNLL use 5 epochs (Appendix Z).

Dataset	n	Greedy	Degen.%	LP ρ	LP Δ	p
CoNLL	2,756	.908	48.4%	.302	-0.85 pp	1.0
SciERC	529	.644	17.8%	.205	$+0.61$ pp	.193
Few-NERD	32,565	.748	13.8%	.220	$+\mathbf{1.39}$ pp	$<.001$

degeneracy (23.4% vs. Qwen 12.7%), while all structural signals degrade selection (VC -2.39 pp; Table 12). Training convergence modulates degeneracy: a 5-epoch Few-NERD control raises degeneracy to 31% and halves LP gains, though selection remains positive (Appendix Z).

Degeneracy alone does not predict LP selection direction. Across 23 dataset-model-temperature configurations at N=8, degeneracy does not predict LP gain (ρ=0.052, p=0.81; Figure 3). Counter-examples illustrate: LLaMA CoNLL exhibits 86.5% degeneracy yet LP is non-negative ($+0.14$ pp), while a 3-epoch SciERC model at 6.4% degeneracy yields LP -1.18 pp. Within fixed settings, however, degeneracy causally modulates LP effectiveness: the SciERC epoch ablation (-1.18, $+0.74$, $+0.99$ pp at 3, 5, 10 epochs; Appendix Z) and per-entity-type analysis within Few-NERD (n=8, $\rho=-0.881$, p=0.004; Appendix X) both show degeneracy effects when dataset and model are held constant. Cross-dataset prediction requires LP signal quality: Few-NERD (within-instance $\rho \approx 0.5$–0.7) shows consistent gains, while SciERC ($\rho \approx 0.26$) and CoNLL ($\rho \approx 0.51$, but only 41% non-degenerate) show mixed results. Even at matched training duration (3 epochs on both datasets), SciERC within-instance ρ=0.319 remains below Few-NERD's $\sim$0.5 (Table 45), confirming that LP-quality alignment, not training duration, drives the cross-dataset difference. All five signals achieve positive QE on Few-NERD (VC ρ=0.39, LP 0.22), yet only LP translates QE into selection; the gap confirms that strong instance-level QE cannot rank candidates within an instance. Conditional analysis on the favorable subset (non-degenerate instances whose within-instance LP range exceeds the dataset-specific median) yields $+2.47$ pp on SciERC (CI $[+0.30, +4.70]$) and a marginal positive trend on Few-NERD ($+0.63$ pp, CI $[-0.02, +1.30]$); CoNLL remains negative (-1.78 pp), consistent with ceiling-regime degeneracy even within the favorable subset (Appendix Z.2).

LP selection scales with sample budget. Doubling N from 8 to 16 on Few-NERD yields $+0.52$ pp (paired p=0.003, CI $[+0.17, +0.87]$), driven by reduced degeneracy (14.6% $\rightarrow$ 7.0%) expanding headroom (12.1 $\rightarrow$ 15.4 pp), while LP

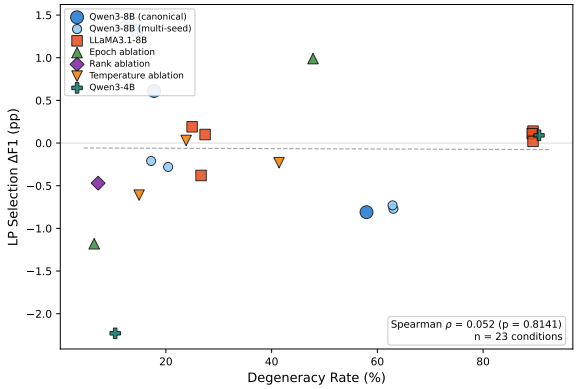


Figure 3: Instance degeneracy vs. LP selection gain across 23 dataset-model-temperature configurations (N=8; ρ=0.052, p=0.81): degeneracy alone does not predict LP direction across datasets. Within Few-NERD per-entity-type analysis (n=8; ρ=−0.881, p=0.004; Appendix X), degeneracy predicts LP gain when LP alignment is held constant. Selection viability depends on LP signal quality and dataset characteristics, not degeneracy alone.

Table 4: Scale ablation: Qwen3-8B vs. 4B NER quality estimation (ρ; N=8, T=1.0, seed=42). 8B values identical to Table 1; 4B uses matched evaluation protocol. RE scale ablation in Appendix L.

		8B	4B	Δ
CoNLL	SJ	.436	.422	−.014
	FK	.400	.418	+.018
SciERC	SJ	.360	.422	+.062
	FK	.267	.326	+.059

ranking quality remains stable (ρ≈0.62). Structural signals remain uniformly negative at N=16; the gap persists regardless of sample budget.

4.5 Robustness

Consistency signals are robust across $T \in \{0.7, 1.0\}$ with sublinear N-scaling gains (Table 8); LP shows the largest temperature sensitivity ($\Delta\rho$=+0.044). Signal rankings are stable across seeds (Kendall's W=1.0 at N=8; m=3 raters, interpret with caution given low m; exact p≈0.028; Table 30). NER calibration is reasonable (ECE 0.069–0.096; Appendix C).

Scale robustness. With Qwen3-4B (Table 4), QE is comparable (mean $|\Delta\rho|$=0.042), yet selection remains below greedy despite 13.0 pp headroom. Task difficulty is a stronger predictor than model scale within the 3B–8B range (Appendix V). Degeneracy scales with capability: 8B shows 17.8% vs. 4B's 10.4% on SciERC (Table 15).

LoRA-rank ablation. Training with LoRA rank=8 (vs. 32) lowers degeneracy from 17.8% to 7.2% and increases headroom to 16.0 pp, but LP does not improve selection (0.635 vs. greedy 0.640). This ablation conflates diversity with model capacity (rank 8 has fewer trainable parameters); within-instance ρ(LP, F1)≈0.24 remains too weak despite wider LP spread (0.057 vs. 0.041 nats). Temperature manipulation (§5) provides a cleaner test by altering diversity without changing capacity, and yields the same conclusion: both low degeneracy and adequate alignment appear independently associated with selection success.

Supervised signal combination. Supervised classifiers (logistic regression, MLP, random forest) trained on all five signals (5-fold CV, 3-seed mean, SciERC, N=16, +17.9 pp headroom) all perform at or below greedy (F1=0.644; LR: −0.4 pp; MLP: −1.6 pp; RF: −3.1 pp). Adding pairwise interaction features does not help (AUROC 0.634 vs. 0.638). To test whether richer representations could close the gap, we train a DeBERTa-v3-base verifier that encodes input-output pairs and predicts per-sample F1. DeBERTa achieves ρ=0.329±0.087 (5-fold CV, 3 seeds). Despite moderate quality prediction, selection utility is poor: on SciERC, DeBERTa closes −26.5±26.5% of the oracle gap (13 of 15 runs negative; vs. LP's +4.2±10.7%); on Few-NERD, DeBERTa recovers only +3.5% (single 80/20 split; vs. LP's +16.3%). Cross-dataset transfer fails entirely (−27% to −72% gap closure). Neither handcrafted signal combinations (LR, MLP, RF on 5 scalar signals) nor learned text representations close the selection gap; only model-internal log-probability provides meaningful selection signal, consistent with an information asymmetry between the generating model and external observers.

4.6 Probing the Information Gap

The preceding negative results suggest the bottleneck is model-internal rather than a feature engineering problem. We test this directly by training probes on the generating model's hidden representations.

Setup. For each of N=8 samples per instance, we extract the mean-pooled final-layer hidden state (d=4096) from the generating Qwen3-8B model and train a linear probe and a two-layer MLP (hidden dim=256) to predict per-sample F1 (5-fold CV, 3 seeds).

Table 5: Hidden state probes vs. LP for sample selection (5-fold CV $\times$ 3 seeds). ρ: Spearman correlation with per-sample F1. Sel. F1: selection F1. Gap%: oracle gap closure. SciERC greedy=0.644, oracle=0.775.

	SciERC			Few-NERD	
Method	ρ	Sel. F1	Gap%	ρ	Gap%
LP	.283	**.650**	+4.8	.606	12.3
Hidden Linear	.179±.019	.613±.002	−23.4	—	
Hidden MLP	**.388±.013**	.637±.003	−5.1	**.629±.007**	11.7

Table 6: Entity-level construction vs. greedy decoding (Δ macro F1; N=8, T=1.0, Qwen3-8B, matching Tables 2–3). LP: LP-weighted entity aggregation; Unif.: frequency-based. $\theta=2/N$ for SciERC; 5-fold CV for others. p: paired bootstrap vs. greedy. Best per dataset in **bold**.

Dataset	Wt.	Greedy	Constr.	Δ	p
SciERC	LP	.644	**.661**	+1.67 pp	.004
	Unif.	.644	.659	+1.54 pp	.023
Few-NERD	LP	.748	**.757**	+0.88 pp	<.001
	Unif.	.748	.751	+0.32 pp	.003
CoNLL	Unif.	.908	.904	−0.39 pp	n.s.

Cross-dataset asymmetry. On SciERC, the MLP probe achieves higher quality correlation than LP (ρ=0.388 vs. 0.283, +37%); hidden states encode information beyond what LP captures. Yet higher correlation does not yield better selection: MLP selection F1 (0.637) falls below LP's (0.650), with gap closure turning negative (−5.1% vs. +4.8%; Table 5). The linear probe performs worse still (ρ=0.179, gap closure −23.4%). On Few-NERD, the MLP provides no increment over LP (gap closure 11.7% vs. 12.3%). A learned token-level LP aggregator (1D CNN over per-token log-probabilities) matches mean LP on Few-NERD (gap closure 56.4% vs. 55.8%), confirming that LP's bottleneck is not token aggregation. The correlation-selection gap thus extends to model-internal representations: in the high-capability regime (greedy F1=0.644), inter-sample F1 differences are small enough that improved ranking provides no marginal selection value, consistent with the capability-diversity tension (§5).

4.7 Prescriptive Results

The preceding sections establish that instance-level selection fails for structured IE. We now present two approaches that extract value from multi-sample generation without relying on sample ranking.

Entity-level construction. Rather than selecting one of N complete outputs, we aggregate entities across all samples. For each unique entity mention, we compute its LP-weighted or uniform frequency across N samples and retain those exceeding a threshold θ (SciERC: $\theta=2/N$; Few-NERD and CoNLL: 5-fold CV; θ=0.5 majority baseline in Appendix Z.4).

LP-weighted construction significantly improves over greedy on SciERC (+1.67 pp, p=0.004) and Few-NERD (+0.88 pp, $p<0.001$; Table 6). CoNLL shows no significant gain (−0.39 pp), consistent with ceiling-regime degeneracy. The mechanism is recall-driven: aggregating low-frequency correct entities across samples recovers mentions that no single sample captures. LP-weighted aggregation outperforms uniform voting at N=8 (SciERC: +1.67 vs. +1.54; Few-NERD: +0.88 vs. +0.32 pp), suggesting that generation probability carries useful information for entity weighting even when it fails for instance-level selection. Increasing N from 8 to 16 further improves construction (Few-NERD: +0.88→+1.32 pp), as more samples provide broader entity coverage; at N=16, LP-weighting and uniform converge, consistent with vote averaging dominating at higher N.

Construction and LP selection are complementary. Construction dominates on SciERC (+1.67 pp vs. LP selection +0.61 pp; Table 2), where LP-quality alignment is weak (within-instance $\rho\approx0.2$). On Few-NERD, LP selection outperforms construction (+1.39 pp vs. +0.88 pp; Table 3), where alignment is strong ($\rho\approx0.5$–0.7). The choice is governed by LP-quality alignment: when LP tracks entity-level F1 within instances, per-sample ranking succeeds; when alignment is weak, entity aggregation bypasses ranking.

LP-alignment gating. A simple gating protocol probes 50 held-out validation instances to estimate within-instance $\rho(\text{LP}, \text{F1})$. Above a threshold, LP selection is applied; below it, entity construction is used instead. This protocol correctly identifies the better method on all three evaluated datasets, and the threshold is stable across $[0.2, 0.4]$. Three caveats apply: (a) in our evaluation, probe instances are drawn from the test set for simplicity; in deployment, a held-out validation set should be used; (b) threshold stability is evaluated on only three datasets; (c) this is a proof of concept that requires broader validation.

Selective prediction. Orthogonal to construction, QE signals support selective prediction: retaining the top-$k\%$ highest-confidence instances. At 80% coverage, selective prediction yields $+5$–8 pp F1 gain (AUPRC 0.70–0.92), with VC as the strongest signal on Qwen (note: VC selection is model-dependent; LLaMA FewNERD VC degrades selection by -2.39 pp, Table 12). For practitioners, when selection fails, QE scores can route low-confidence instances to human review rather than sample N candidates to identify the best output.

5 Discussion

Implications. All signals achieve positive quality estimation across non-collapsed datasets, with rankings stable across model families and a 4B ablation (Table 4; RE in Appendix L). For selection, LP gains are associated with strong LP-quality alignment (Table 3); structural signals uniformly degrade selection. The decoupling of ρ, AUROC, and selection F1 means single-metric evaluation risks selecting the wrong signal; effective selection likely requires content-aware verifiers (Cobbe et al., 2021; Venktesh et al., 2025).

Why the gap exists. Two factors constrain selection: *instance degeneracy* (consensus-proximity) limits the available headroom, and *weak LP-quality alignment* (per-sample) prevents exploiting the remainder. Within non-degenerate instances, MBR utility (Eikema and Aziz, 2022) correlates positively with F1 (SJ median ρ=0.524), but 17.8% (SciERC) to 48.4% (CoNLL) of instances produce identical F1 across all samples, eliminating headroom entirely (Table 15). LP is severely compressed (median range 0.041 nats; Table 19), with within-instance $\rho(\text{LP}, \text{F1})$ ranging from 0.262 (SciERC) to 0.667 (Few-NERD; Table 23). Where compression is mitigated (LP range $\geq$0.05, n=210), within-instance ρ rises to 0.39 and LP selection yields $+2.9$ pp (p=0.004; Table 22). An oracle tiebreaker within VC-equivalent groups recovers $+7.9$ pp, confirming per-sample signal exists but current signals cannot extract it (Appendices R, U, K). Hidden state probes confirm richer information exists (SciERC MLP ρ=0.388 vs. LP 0.283, $+37\%$), but the correlation-selection gap persists (gap closure -5.1% vs. $+4.8\%$; §4.6): the bottleneck is diminishing marginal returns on selection, not information loss.

LP compression mechanisms. Four non-mutually-exclusive mechanisms contribute (Appendix P); entity-token LP shows $4\times$ higher variance yet performs worse; compression operates at the representation level, not token aggregation (Appendix Z.3).

Regime boundaries. QE signals collapse at the low-resource boundary (WNUT-17, conditional ρ=0.038; Table 11) and LP degrades selection at the ceiling (CoNLL, degeneracy 48–87%, -0.85 pp). Temperature manipulation disentangles the two factors: at T=1.2, degeneracy drops to 14.9% yet LP does not improve because alignment remains weak (ρ=0.225 vs. Few-NERD $\sim$0.5–0.7; Table 35); the SciERC epoch curve corroborates this (Table 45). Within fixed settings, degeneracy causally modulates LP effectiveness: the SciERC epoch ablation shows LP gains varying with training-induced degeneracy (Table 45). Across 23 dataset-model-temperature configurations (N=8), however, degeneracy alone does not predict LP direction (ρ=0.052, p=0.81); LP signal quality and dataset characteristics are required for cross-dataset prediction. Temperature and training convergence are controllable levers for degeneracy (Tables 35, 44), while calibration cannot improve selection (Appendix Y).

Capability-diversity tension. Degeneracy rises with model scale (4B: 10.4% vs. 8B: 17.8%; Table 4), training convergence (Few-NERD: 14% at 3 epochs to 31% at 5; Tables 3, 44, 45), and model family (LLaMA 86.5% vs. Qwen 48.4% on CoNLL; LLaMA 23.4% vs. Qwen 12.7% on Few-NERD; Table 12). For structured IE, the same training progress that reduces error also reduces the output diversity on which selection depends (Wang et al., 2023; Snell et al., 2025). This pattern is observational and does not establish causality, but the consistency across independent axes suggests a systematic relationship: as models improve, the window for test-time selection narrows.

From diagnosis to practice. Our diagnostic motivates a shift from instance-level selection to entity-level construction (§4.7). A degeneracy-aware LP strategy (DALS; Appendix Z.1) and MBR selection both fail to reliably beat greedy; degeneracy-gated selection (DGS) shows mixed results (Few-NERD $+0.92$ pp, SciERC $+0.71$ pp non-significant, CoNLL sign-reversal across models; Appendix Z.4). Entity-level construction by-

passes both failure mechanisms: it aggregates across samples rather than ranking them, recovering $+1.67$ pp (SciERC) and $+0.88$ pp (Few-NERD) via LP-weighted entity aggregation (Table 6). Construction and LP selection are complementary, governed by LP-quality alignment (§4.7), and a simple alignment-based gating protocol selects the better method per dataset. For practitioners, QE-based selective prediction provides an orthogonal, immediate payoff: routing low-confidence instances to human review yields $+5$–8 pp at 80% coverage without requiring multi-sample generation at inference time.

When would consistency-based selection work? Our negative results are specific to fine-tuned models with constrained decoding on structured IE. Several conditions outside our experimental scope could restore selection headroom. (a) Models with poor calibration or in-context learning setups (without fine-tuning) produce more diverse outputs, widening the gap between candidates that consistency signals can exploit. (b) Closed-source APIs where log-probabilities are unavailable make consistency signals the only viable selection mechanism; our structural signals (SJ, VC, FK, EM) provide quality estimation (ρ up to 0.45) even without model-internal access. (c) Ensemble or mixture-of-experts settings generate structurally diverse candidates by design, reducing instance degeneracy below the levels observed in our single-model sampling. (d) Tasks where the model exhibits systematic overconfidence (e.g., hallucination-prone generation) may benefit from consensus filtering, as disagreement among samples would carry more discriminative signal than in our well-calibrated fine-tuned regime.

6 Conclusion

Five confidence signals achieve reliable NER quality estimation (ρ up to 0.45, AUROC up to 0.82), enabling selective prediction ($+5$–8 pp F1 at 80% coverage). Instance-level selection fails due to degeneracy and LP compression, but the correct self-consistency analogy for structured outputs is entity-level construction: LP-weighted entity aggregation yields $+1.67$ pp on SciERC ($p{=}0.004$) and $+0.88$ pp on Few-NERD ($p{<}0.001$). Construction and LP selection are complementary, governed by LP-quality alignment; a alignment-based gating protocol routes the appropriate method per dataset. For practitioners: use QE-based selective prediction when deployment requires reliability guarantees; apply entity construction when multi-sample generation is feasible; assess LP-alignment via held-out validation to choose between construction and LP selection. As models improve, degeneracy rises and the selection window narrows; content-aware verifiers (Cobbe et al., 2021) may be needed to close the remaining gap.

Limitations

Our evaluation covers four NER datasets; an exploratory RE extension on SciERC is reported in Appendix L but generalization to other RE benchmarks remains unvalidated. We cover 4B–8B scale; whether the gap holds for 70B+ models or closed-source APIs is untested. Larger models may produce more diverse samples (potentially reducing degeneracy), but their LP-quality alignment behavior is unknown, and these two factors may shift in opposite directions. Our supervised approaches (DeBERTa verifier, hidden state probes, MLP/RF on signal combinations) all fail to close the oracle gap; whether content-aware verifiers with access to task-specific structure could succeed remains open.

We evaluate at $N{=}8$ (primary) and $N{=}16$; behavior at larger sample sizes is unknown (Appendix R). The free-form ablation covers SciERC NER and CoNLL NER with one model; RE free-form results are in Appendix L. Qwen3-8B's SciERC checkpoint used empty `<think></think>` blocks; a three-part analysis rules out this confound (Appendix I). LP compression worsens on easy datasets at lower temperature (CoNLL tied fractions reach 95.8% at $T{=}0.7$); whether reduced temperature translates to selection gains remains untested. We tested verbalized confidence (Lin et al., 2022; Tian et al., 2023) as a second per-sample signal; it exhibits severe overconfidence (mean 8.58/10 on SciERC, 9.24/10 on CoNLL) and yields the weakest QE and selection among all signals (Appendix U). Ad-hoc LP$\times$VC combinations yield at most $+0.2$ pp over greedy (not significant; §5); whether *learned* combinations could close the gap remains open. The $\varepsilon{=}0.05$ threshold used for tied/non-tied LP stratification is a post-hoc exploratory choice; while selection F1 is stable across $\varepsilon \in [0, 0.2]$ (Appendix J), the specific subset gains should be interpreted as characterizing the compression phenomenon rather than as a tuned selection strategy. We tested one remedy for LP compression (entity-token-only LP, excluding for-

mat/schema tokens) which fails to improve selection despite increasing within-instance variance on SciERC ($4\times$); LP$\times$VC combinations also fail to recover selection (§5). Our supervised signal combination baseline uses only the five unsupervised signals studied; a DeBERTa verifier encoding input-output pairs also fails to close the gap (preliminary, SciERC n=106), hidden state probes on the generating model's representations improve correlation ($+37\%$ on SciERC) but do not improve selection (gap closure -5.1%; §4.6), confirming the bottleneck extends to model-internal access. Entity embeddings and calibration-based approaches remain untested. Other per-sample scoring methods exist but were not evaluated: Self-Certainty (Kang et al., 2025) (consistency of token-level self-evaluation) and P(True) (Kadavath et al., 2022) (model's own probability estimate of correctness) both produce per-sample scores and could in principle address the alignment bottleneck; whether they suffer from similar compression under fine-tuning is unknown. Our selection experiments use 3 training seeds for Few-NERD and CoNLL (stability $\sigma \leq 0.04$ pp for LP selection), confirming robustness on the key positive and negative datasets; SciERC uses 1 primary seed with high seed sensitivity (σ=0.48 pp, n=529), reflecting limited within-seed statistical power. Few-NERD models are trained for 3 epochs (vs. 5 for other datasets) due to earlier convergence; epoch confound analyses on both Few-NERD (2-point) and SciERC (3-point, including 10-epoch overfitting) are provided in Appendix Z.

Ethics Statement

All experiments use publicly available benchmarks (SciERC, CoNLL-2003, WNUT-17) and open-weight models (Qwen3-8B, Qwen3-4B, LLaMA-3.1-8B). CoNLL-2003 is derived from the Reuters RCV1 corpus, which is available for research purposes under a NIST agreement; we use the standard research distribution. No new human data was collected or annotated for this study. Our findings on signal limitations could inform more reliable deployment of IE systems by guiding practitioners toward appropriate confidence signals for their use case. Code and data will be released upon acceptance at [anonymized URL].

Responsible NLP Research Checklist

A For Every Submission

A1. **Did you discuss the limitations of your work?**
Yes. See the Limitations section after the Conclusion.

A2. **Did you discuss any potential risks of your work?**
Yes. See the Ethics Statement. This work analyzes existing confidence signals for NER and identifies when they fail; the findings reduce risk of deploying unreliable confidence-based selection in production systems. No dual-use concerns.

A3. **Do the abstract and introduction summarize the paper's main claims?**
Yes. The Abstract (4 claims) and Section 1 (4 enumerated contributions) summarize all main claims with supporting evidence pointers.

B Did you use or create scientific artifacts?

Yes. We use existing datasets and pre-trained models.

B1. **Did you cite the creators of artifacts you used?**
Yes. All datasets are cited: SciERC (Luan et al., 2018), CoNLL-2003 (Sang and Meulder, 2003), Few-NERD (Ding et al., 2021), WNUT-17 (Derczynski et al., 2017). Models: Qwen3-8B/4B and LLaMA-3.1-8B-Instruct. Training framework: LLaMA Factory (Zheng et al., 2024). Constrained decoding: XGrammar (Dong et al., 2024). See Sections 1, 3.3, and Appendix H.

B2. **Did you discuss the license or terms for use and/or distribution of any artifacts?**
Partially. The Ethics Statement notes that CoNLL-2003 is derived from the Reuters RCV1 corpus, available under a NIST agreement for research. SciERC, Few-NERD, and WNUT-17 are publicly available research benchmarks. Qwen3 and LLaMA-3.1 are open-weight models released under their respective licenses (Qwen License, Llama 3.1 Community License). Code and data will be released upon acceptance.

B3. **Did you discuss if your use of existing artifact(s) was consistent with their intended use?**
Yes. All datasets are used for their intended NER evaluation purpose. Models are used for fine-tuning and inference on NER, consistent with their general-purpose design. See Ethics Statement.

B4. **Did you discuss the steps taken to check whether the data contains any information that names or uniquely identifies individual people or offensive content?**
N/A. We use established, widely-studied NER benchmarks (news articles, scientific abstracts, social media). No new data was collected or annotated.

B5. **Did you provide documentation of the artifacts?**
Yes. Dataset domains and characteristics are described in Section 3.3. Training configuration details are in Appendix H.

B6. **Did you report relevant statistics like the number of examples, details of train/test/dev splits, etc.?**
Yes. Instance counts are reported in Table 3 and throughout Section 4. Gold-filtering counts, dataset sizes, and evaluation protocol details are in Section 3.3.

C Did you run computational experiments?

Yes. See Section 4.

C1. **Did you report the number of parameters in the models used, the total computational budget, and computing infrastructure used?**
Partially. Model sizes (4B, 8B parameters) are stated in Section 1. LoRA configuration and training hyperparameters are in Appendix H. Computing infrastructure: 4× NVIDIA RTX 5090 GPUs. We do not report total GPU hours; each model fine-tunes in under 2 hours and inference for $N{=}8$ samples across all test sets completes in under 1 hour per configuration.

C2. **Did you discuss the experimental setup, including hyperparameter search and best-found hyperparameter values?**
Yes. Appendix H (Table 18) reports all hyperparameters: LoRA rank/alpha, learning rate, batch size, epochs, optimizer, precision, etc. Both models use identical settings to ensure fair comparison. No extensive hyperparameter search was performed; we use standard LoRA fine-tuning defaults. LoRA rank is ablated in Section 4.5.

C3. **Did you report descriptive statistics about your results?**
Yes. We report multi-seed results (3 seeds; σ in Table 17), bootstrap 95% confidence intervals (Tables 2, 39), paired bootstrap p-values throughout, and minimum detectable effect sizes (Section 4.3). Tables clearly indicate whether values are single-seed, multi-seed means, or confidence intervals.

C4. **If you used existing packages, did you report the implementation, model, and parameter settings used?**
Yes. LLaMA Factory (Zheng et al., 2024) for fine-tuning and XGrammar (Dong et al., 2024) for constrained decoding are cited with version-relevant details in Appendix H. Evaluation uses standard span-level micro-F1 (converted to per-instance macro-F1 as described in Section 3.3).

D Did you use human annotators or research with human subjects?

No. All experiments use existing annotated benchmarks. No new human annotation or human subjects research was conducted.

E AI Assistants

E1. **Did you include information in your paper about your use of AI assistants?**
Yes. AI writing assistants were used for drafting and editing. All scientific claims, experimental results, and analyses are the authors' own work, verified against experimental outputs.

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A Notation

Table 7 summarizes the notation used throughout the paper.

Symbol	Meaning
$\mathcal{M}$	Language model
x	Input instance
y_i	i-th stochastic sample
N	Number of samples
T	Sampling temperature
$c(\cdot)$	Confidence signal
SJ	Soft Jaccard similarity
FK	Fleiss' κ
EM	Exact-match voting rate
VC (c_{vc})	Entity voting confidence
LP (c_{lp})	Log-probability (mean token log-likelihood)

Table 7: Notation used throughout the paper.

B Robustness Analysis

We exclude instances with empty gold annotations from all correlation analyses (CoNLL: 697/3,453 instances, 20.2%; SciERC: 22/551, 4.0%). For these instances, F1 is trivially zero regardless of prediction quality, and consistency signals reflect agreement on the empty set rather than extraction quality. Including them inflates or deflates Spearman ρ depending on signal behavior at the boundary.

Temperature sensitivity. SJ NER ρ shifts from 0.364 (T=0.7) to 0.360 (T=1.0), a difference of 0.004. FK NER ρ drops by 0.022, while RE ρ increases by 0.091, likely reflecting reduced diversity at lower temperature interacting differently across tasks. LP exhibits the largest temperature effect on NER (ρ=0.249 at T=0.7 vs. 0.205 at T=1.0, Δ=+0.044), consistent with lower temperature concentrating probability mass and improving LP's discriminative power for quality estimation; RE T=0.7 LP data is not available. Across all comparisons, the signal hierarchy is preserved at both temperatures on NER (SJ>FK>LP at T=0.7: 0.364>0.289>0.249; at T=1.0: 0.360>0.267>0.205).

Table 8: Signal robustness across configurations (Spearman ρ). Top: temperature on Qwen SciERC. Middle: sample size on Qwen SciERC. Bottom: transfer across model families (both 8B) (LLaMA CoNLL N=16, 3-seed mean $\pm\sigma$). †Mean over 3 seeds (σ=0.013; individual: 0.398, 0.423, 0.410). σ computed as sample standard deviation.

		NER ρ	RE ρ
Temperature (Qwen SciERC, N=8)			
SJ	T=0.7	0.364	0.234
SJ	T=1.0	0.360	0.250
FK	T=0.7	0.289	0.184
FK	T=1.0	0.267	0.275
LP	T=0.7	0.249	—
LP	T=1.0	0.205	0.266
Sample size (Qwen SciERC, T=1.0)			
SJ	N=8	0.360	0.250
SJ	N=16	0.411†	0.281
FK	N=8	0.267	0.275
FK	N=16	0.283	0.279
Across model families (LLaMA CoNLL, N=16, 3 seeds)			
SJ		0.474±0.009	—
FK		0.467±0.009	—
EM		0.473±0.010	—
VC		0.470±0.006	—
LP		0.311±0.003	—

Sample size scaling. Increasing from N=8 to N=16 improves SJ on Qwen SciERC NER: ρ rises from 0.360 to 0.411±0.010 (three seeds: 0.398, 0.423, 0.410), a gain of +0.047. The magnitude of N-scaling gains is setting-dependent (LLaMA SciERC NER yields +0.016 and Qwen CoNLL NER +0.014), though the direction is consistently positive. LLaMA CoNLL exhibits the most uniform scaling: all four consistency signals gain $\Delta\rho\approx$+0.042 (range: +0.039 to +0.044) while LP gains only +0.004 (Table 1). FK N-scaling is seed-dependent: Qwen SciERC gains range from +0.005 to +0.023 across seeds (mean +0.016), while LLaMA CoNLL yields +0.039 (3-seed mean); individual seed FK shifts vary substantially under the unified pipeline, so this should not

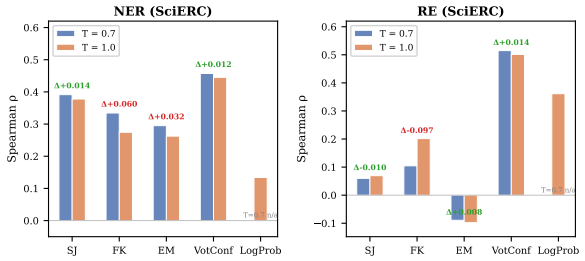


Figure 4: Temperature robustness on SciERC (N=8). Top: NER; bottom: RE. SJ maintains stable ρ across T=0.7 and T=1.0, while FK shows mixed sensitivity. LP shows the largest NER temperature effect ($\Delta\rho$=+0.044); RE T=0.7 LP data unavailable.

be interpreted as a robust trend. VC N-scaling is consistently positive (+0.005 on Qwen CoNLL, +0.018 on LLaMA SciERC, +0.030 on Qwen SciERC), while LP is approximately N-invariant ($\Delta \leq 0.034$). SJ, EM, and VC scale consistently across all tested settings; only FK shows seed-dependent variation that does not constitute a reliable trend, structural and binary-agreement signals are the most robust to N. At N=16, SJ converges to comparable NER performance with VC (Δ=0.002) while exhibiting lower cross-seed variance (σ=0.010 vs. 0.026), a more stable alternative at higher sample sizes.

Transfer across model families. LLaMA-3.1-8B-Instruct shows a similar SJ–FK ordering on SciERC NER (SJ ρ=0.325±0.021, FK ρ=0.256±0.024, $\Delta\rho$=0.069; permutation p=0.06, borderline). On CoNLL, the gap across model families is minimal (Table 1): SJ achieves ρ=0.431 (LLaMA) vs. 0.436 (Qwen), Δ=0.005. At N=16, the structural-over-LP advantage holds across all four tested NER settings: Qwen SciERC (VC ρ=0.409 vs. LP 0.239, Δ=+0.170), Qwen CoNLL (EM ρ=0.463 vs. LP 0.328, Δ=+0.135),[5] LLaMA SciERC (VC ρ=0.452 vs. LP 0.277, Δ=+0.175), and LLaMA CoNLL 3-seed (SJ ρ=0.474±0.009 vs. LP 0.311±0.003, Δ=+0.163).

FK independence: instance-level permutation test. To verify that Fleiss' κ reflects genuine within-instance agreement rather than marginal-rate artifacts, we conduct instance-level permutation tests across three configurations. For each instance, we permute sample labels 10,000 times to generate a null distribution of FK under independence, then compare the observed FK against this null via a one-sample t-test and permutation p-value.

[5] Qwen CoNLL N=16 was not recomputed under the unified pipeline; LP ρ=0.328 may differ slightly from the registry value (0.315) due to instance filtering.

Table 9: Instance-level FK independence test. Permutation tests confirm FK reflects true within-instance agreement (all p<0.001). Split-half reliability r computed by randomly partitioning N samples into two halves and correlating their FK scores.

Config	n	FK mean	95% CI	t	Split-half
SciERC (Qwen)	529	0.402	[0.376, 0.428]	30.8	0.395
CoNLL (LLaMA)	2,756	0.927	[0.919, 0.935]	225.9	0.578
CoNLL (Qwen)	2,756	0.720	[0.706, 0.734]	102.1	0.520

Table 9 reports the results. All configurations reject the independence null at p<0.001 (n=529–2,756), FK captures genuine inter-sample agreement rather than chance label co-occurrence. Split-half reliability ranges from r=0.395 (SciERC) to r=0.578 (CoNLL LLaMA), moderate stability: FK estimates from half the samples correlate positively with those from the other half, though the SciERC value suggests noisier estimation on harder tasks with fewer samples. The higher FK means on CoNLL (0.720–0.927) vs. SciERC (0.402) mirror the difficulty gradient observed in the main quality estimation results.

C Calibration Analysis

Reliability diagrams use 10 equal-width bins. NER signals are reasonably well-calibrated (calibration ECE 0.069–0.096), while RE exhibits greater miscalibration (ECE 0.109–0.339), translating consistency scores into calibrated confidence estimates is feasible for NER but remains challenging for compound extraction tasks.

D Additional Analyses

Per-relation diagnostic. RE signal degradation is relation-specific. FEATURE-OF has near-zero correlation (ρ=0.004), while HYPONYM-OF achieves ρ=0.646. The primary bottleneck is entity span errors, which account for 55.5% of RE false negatives. This is a prerequisite failure that consistency signals cannot detect, as they measure agreement among equally incorrect extractions. We note that RE analysis relies on a single dataset (SciERC); generalization to other RE benchmarks remains to be validated.

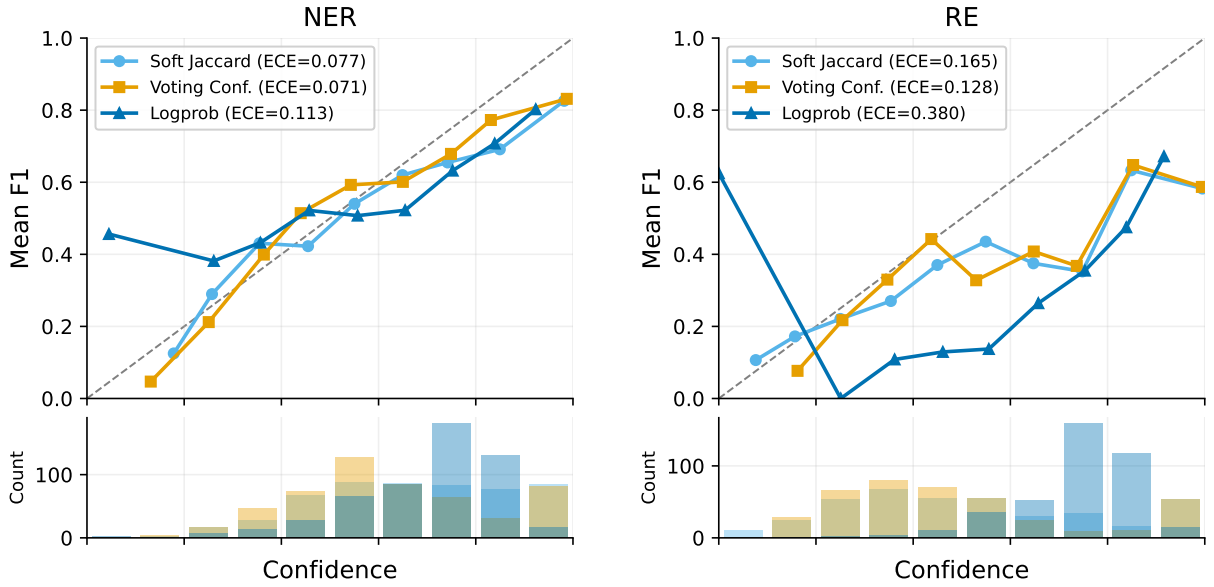


Figure 5: Reliability diagrams on SciERC (N=8, T=1.0). Top: mean F1 vs. binned confidence; bottom: instance counts per bin. Left: NER; right: RE. NER signals are reasonably well-calibrated (calibration ECE 0.069–0.096), while RE shows greater miscalibration (ECE 0.109–0.339), RE calibration remains challenging for compound extraction tasks.

Capability stratification. For NER, SJ effectiveness scales with structural complexity: on SciERC instances with 6+ entities, ρ=0.547, compared to ρ=0.415 for 1–2 entities. The SJ advantage over LP remains stable across sentence-length quartiles ($\Delta\rho\approx$0.08–0.20). For RE, SJ ρ increases monotonically with sentence length (Q1: 0.110 $\rightarrow$ Q4: 0.473; Fisher z-test p=0.004): longer, more relation-dense sentences provide richer structure for consistency-based estimation. Both proxies are weakly correlated with F1 ($|r|<0.15$), they capture input complexity rather than directly predicting output quality; however, entity count and sentence length are themselves moderately correlated (ρ=0.650), so they reflect overlapping rather than independent complexity dimensions. For RE, entity count and relation count are strongly correlated (ρ=0.769) and thus reflect a single structural complexity dimension rather than independent axes. On the high-complexity RE subset (n=81, high relation count), signals improve substantially: VC reaches ρ=0.630, LP ρ=0.554, and SJ ρ=0.502, all far above the full-set levels. Richer relational structure provides more signal for consistency-based estimation, complementing the task-level boundary established by WNUT-17: structural signals are most effective within a "sweet spot" of sufficient model capability and sufficient output complexity.

Length vs. log-probability. Figure 7 shows that mean log-probability is weakly positively correlated with output length (NER: ρ=0.386, n=4,232; RE: ρ=0.326, n=3,176), ruling out length-penalty confounding as an explanation for LP's selection failure: longer outputs receive *higher*, not lower, token-averaged log-probability under constrained decoding.

Evaluation circularity and LOO-SJ equivalence. Consensus-proximity selection chooses $\hat{y} = \arg\max_{y_i} \sum_{j \neq i} \mathrm{SJ}(y_i, y_j)$. Leave-one-out SJ (LOO-SJ) for sample y_i is $\text{LOO-SJ}_i = \frac{1}{N-1} \sum_{j \neq i} \mathrm{SJ}(y_i, y_j)$. Since the $\frac{1}{N-1}$ constant is shared across all candidates within an instance, $\arg\max_{y_i} \text{LOO-SJ}_i = \arg\max_{y_i} \sum_{j \neq i} \mathrm{SJ}(y_i, y_j)$; the two selection rules are equivalent. Standard (non-LOO) SJ adds the self-comparison term $\mathrm{SJ}(y_i, y_i) = 1$ to each candidate, which is constant and thus does not affect ranking. Empirically, LOO-SJ yields $\Delta\rho = -0.014$ relative to standard SJ (a correlation-level difference), with a maximum per-instance score difference of 3.1×10^{-4} (negligible at the individual prediction level). Evaluation circularity is negligible.

SJ normalizer robustness. Equation 4 uses $\max(|g_i|, |g_j|)$ as the denominator; an alternative is the union size $|g_i \cup g_j|$. Table 10 compares the

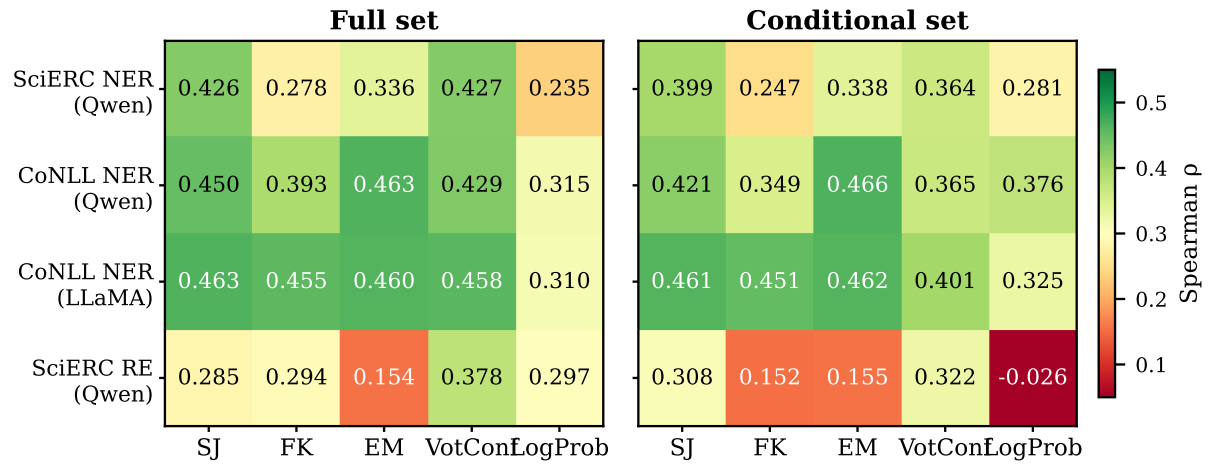


Figure 6: Quality estimation heatmap (Spearman ρ) across five signals and four configurations. Left: full evaluation set; right: conditional set (non-zero F1 instances). On the full set, structural signals dominate on SciERC NER while all signals converge on CoNLL. On the conditional set, EM emerges as the strongest signal on both NER (ρ=0.360) and RE (ρ=0.417), while LP RE collapses to near-zero (ρ=0.011).

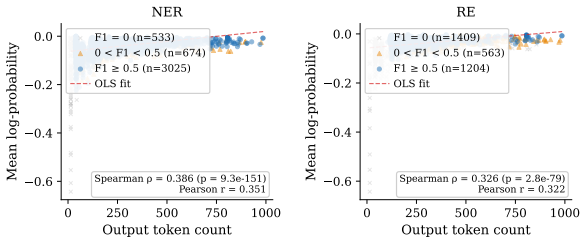


Figure 7: Output length vs. mean log-probability on SciERC (N=8, T=1.0). Positive correlations (NER: ρ=0.386; RE: ρ=0.326) rule out length-penalty confounding.

two normalizers across all six model–dataset combinations (N=8, T=1.0, seed=42): $|\Delta\rho|<0.01$ in every case, the choice between max and union normalization does not affect conclusions.

Table 10: SJ normalizer comparison (Spearman ρ; N=8, T=1.0, seed=42). Max: $\max(|g_i|, |g_j|)$; Union: $|g_i \cup g_j|$.

Dataset	Model	ρ(max)	ρ(union)	$\Delta\rho$
CoNLL	Qwen3-8B	.436	.438	−.002
CoNLL	Qwen3-4B	.422	.421	+.000
CoNLL	LLaMA-8B	.431	.432	−.000
SciERC	Qwen3-8B	.360	.365	−.005
SciERC	Qwen3-4B	.422	.423	−.001
SciERC	LLaMA-8B	.363	.372	−.009

Conditional filtering details. On LLaMA conditional RE (non-zero F1 instances), SJ recovers to ρ=0.405 while FK drops to ρ=0.259, consistent with the NER structural advantage; on full-set RE, however, the SJ–FK difference is non-significant ($\Delta\rho$=−0.005, p=0.53; preliminary). At N=16, the divergence persists: SJ maintains ρ=0.308 on conditional RE while LP drops to ρ=−0.026. EM ranks first across all four tested conditional configurations: SciERC NER (ρ=0.360), CoNLL Qwen N=16 (ρ=0.466), CoNLL LLaMA 3-seed (ρ=0.462), and SciERC RE (ρ=0.417). This holds despite full-set rankings ranging from #1 (CoNLL Qwen) to #5 (SciERC RE). On the full set, EM is diluted by trivial agreement: instances where all N samples produce identical (possibly incorrect) outputs receive EM=1 regardless of quality, compressing the signal's dynamic range. On the conditional set, where the model demonstrably produces varied outputs, exact agreement becomes the strongest quality signal. The margin over SJ is largest on RE ($\Delta\rho$=+0.171), where compound extraction decisions amplify output variability, and smallest on CoNLL LLaMA ($\Delta\rho$=+0.002), where all signals converge near $\rho\approx$0.46.

WNUT-17 full signal analysis. Table 11 reports all five signals on WNUT-17 with full-set and conditional ρ.

The full-set p-values are: VC <1e-112, FK p=0.001, SJ p=0.020, EM p=0.194, LP <1e-28. VC's high full-set ρ=0.724 is entirely driven by the zero/non-zero boundary (conditional ρ=0.038, p=0.49), while LP's conditional ρ=0.350 is the highest among all signals on the non-zero subset, full-set correlations on high-failure-rate datasets conflate binary success prediction with quality gradation.

Table 11: Quality estimation on WNUT-17 (Qwen, N=16, T=1.0, n=689). Cond: non-zero F1 instances (n=334). Full-set ρ reflects binary success prediction; conditional ρ measures quality gradation. LP's conditional ρ is highest despite negative full-set ρ.

Signal	ρ	AUROC	Cond. ρ	Cond. p
VC	.724	.913	.038	.487
FK	.125	.555	.169	.002
SJ	.089	.517	.288	<1e-7
EM	.050	.487	.333	<1e-9
LP	−.410	.219	.350	<1e-10

Cross-configuration selection analysis. Table 12 extends the selection analysis to five additional configurations. On LLaMA SciERC, where output diversity remains substantial (18% identical outputs, 39% unique), LP achieves the highest selection F1 (0.652) at N=8 but still falls within bootstrap noise of single-sample (0.646), replicating the gap observed on Qwen SciERC (Table 2). On LLaMA FewNERD (n=32,565, 3-seed mean), LP is the only signal with positive selection (+0.36 pp), confirming LP as the most robust selection signal across model families (Qwen FewNERD: +1.39 pp, Table 3). All structural signals degrade selection (VC −2.39 pp, SJ −0.52 pp), consistent with the Qwen pattern. LLaMA FewNERD degeneracy (23.4±4.7%, 3 seeds) is higher than Qwen (12.7%), illustrating the capability-diversity tension: LLaMA achieves stronger greedy F1 (0.784 vs. 0.748, +3.6 pp) but at the cost of higher degeneracy (+10.7 pp) and reduced oracle headroom (10.0 vs. 12.4 pp). On CoNLL, 48–87% of instances produce identical outputs across all N samples; this reflects a ceiling effect (single-sample F1 ≈0.91–0.92) rather than signal failure, and selection analysis on these configurations is uninformative because there is little per-sample variance for any signal to exploit.

Diminishing returns of N-scaling. Our N=8 and N=16 comparisons (Table 8) characterize the marginal value of additional samples for test-time scaling (Brown et al., 2024; Snell et al., 2025). For quality estimation, structural signals benefit consistently but sublinearly: SJ ρ improves by +0.047 (Qwen SciERC, 3-seed mean) and +0.032 (LLaMA CoNLL, 3-seed mean) when doubling from N=8 to N=16, while log-probability is approximately N-invariant ($\Delta\rho \leq 0.021$). Structural signals use $\binom{N}{2}$ pairwise comparisons (28 pairs at N=8, 120 at N=16), but the ρ improvement is far less than proportional, each additional comparison yields diminishing information. This asymmetry gives structural signals a relative advantage in the compute-constrained regime: at N=8, SJ already achieves ρ=0.431 on LLaMA CoNLL (vs. LP ρ=0.307, Δ=+0.124), and the gap *widens* at N=16 (Δ=+0.163) because structural signals continue to benefit from additional pairwise evidence while LP does not. For sample selection, additional samples increase the oracle upper bound (more draws raise the probability of a high-quality candidate), but signal-based selection remains flat because the selector's inability to identify the best candidate is a granularity limitation, not a sample size limitation. The practical implication is that N=8 captures most of the achievable quality estimation benefit; further scaling widens rather than closes the correlation-selection gap absent stronger per-sample selectors.

N=16 CoNLL selection: small but significant. While quality estimation benefits sublinearly from N-scaling, selection behavior at N=16 reveals a nuanced exception on LLaMA CoNLL. VC achieves the largest gain over single-sample (+0.28 pp, p=0.003–0.027 across 3 seeds, all significant), with SJ, FK, and EM at +0.22–0.23 pp and LP at +0.12 pp (2/3 seeds significant). These gains are statistically significant but practically minimal: oracle headroom at N=16 is only 2.5 pp, so VC captures approximately 10% of the available headroom. This reinforces the primary finding: even under the most favorable conditions (high baseline performance, more candidates, and aggregated multi-seed evidence), signal-based selection recovers only a fraction of the oracle headroom available.

E Bootstrap CI for Signal Comparisons

Table 13 reports bootstrap confidence intervals for individual signal ρ and pairwise $\Delta\rho$ on SciERC NER (N=8, n=529, B=10,000, seed=42). The SJ–LP gap ($\Delta\rho$=+0.155) is clearly significant, the structural advantage reported in §4.2 is robust. The VC–SJ gap ($\Delta\rho$=+0.019) is not significant; these two structural signals perform comparably. The SJ–EM gap ($\Delta\rho$=+0.065) is borderline significant, with the CI lower bound at +0.003 and upper bound at +0.130.

At N=16 (B=10,000, permutation test), the structural advantage extends to FK comparisons. On SciERC NER (n=529), SJ–FK ($\Delta\rho$=0.147–

Table 12: Best-of-N selection F1 (macro, instance-averaged) across configurations (T=1.0, 3-seed mean). Best non-oracle signal per row in **bold**. LP is the only signal with positive selection on both FewNERD configurations (Qwen +1.39 pp, Table 3; LLaMA +0.36 pp). At N=16 on LLaMA CoNLL, VC achieves small but significant gains (+0.28 pp, p<0.03; see text below). CoNLL shows especially tight clustering where greedy already exceeds 0.90.

Config	n	Greedy	Oracle	SJ	FK	EM	LP	VC
LLaMA/SciERC N=8	529	.646	.778	.643	.639	.634	**.652**	.633
LLaMA/SciERC N=16	529	.645	.806	**.649**	**.649**	.640	.646	.640
LLaMA/FewNERD N=8	32,565	.784	.884	.779	.779	.774	**.788**	.760
Qwen/CoNLL N=8	2,756	.908	.957	**.899**	.893	.898	.897	.895
LLaMA/CoNLL N=8	2,756	.924	.943	.924	**.925**	.924	.924	.923

Table 13: Bootstrap 95% CIs for signal ρ on SciERC NER (N=8, n=529, B=10,000, seed=42). Top: individual signals. Bottom: pairwise $\Delta\rho$; "Sig." indicates whether the 95% CI excludes zero.

Signal	ρ	95% CI
VC	+0.379	[+0.299, +0.456]
SJ	+0.360	[+0.277, +0.437]
EM	+0.295	[+0.212, +0.371]
FK	+0.267	[+0.176, +0.354]
LP	+0.205	[+0.119, +0.291]

Comparison	$\Delta\rho$	95% CI	Sig.
SJ − LP	+0.155	[+0.074, +0.234]	Yes
VC − SJ	+0.019	[−0.033, +0.073]	No
SJ − EM	+0.065	[+0.003, +0.130]	Marginal[†]

[†]Marginally significant without multiple comparison correction; would not survive Bonferroni correction for 10 pairwise comparisons (α_{adj}=0.005).

Table 14: Output diversity across configurations (N=8, T=1.0). "Identical" = fraction of instances with all N outputs identical. "Unique" = mean fraction of distinct outputs per instance.

Config	Identical	Unique
Qwen/SciERC	11%	43%
LLaMA/SciERC	18%	39%
Qwen/CoNLL	57%	—
LLaMA/CoNLL	88%	—
WNUT-17	55%	—

0.160, all $p \leq 0.002$) and VC–FK ($\Delta\rho$=0.138–0.188, all $p \leq 0.003$) are clearly significant across all three seeds; EM–FK is seed-dependent ($\Delta\rho$=0.054–0.093, p=0.048–0.15). On CoNLL (n=2,756), all signals significantly outperform FK: EM $\Delta\rho$=0.071 (p<0.001), SJ $\Delta\rho$=0.056 (p<0.001), VC $\Delta\rho$=0.022 (p=0.040).

F Diversity and Selection Analysis

Output diversity quantification. Table 14 summarizes output diversity across configurations. On SciERC, output diversity is substantial: Qwen produces 43% unique outputs per instance (mean 3.5 distinct out of N=8), with per-sample F1 standard deviation of 0.144 and mean pairwise SJ of 0.703. LLaMA shows similar diversity (39% unique, F1 std 0.11). On CoNLL, diversity collapses: 59% (Qwen) to 89% (LLaMA) of instances produce identical outputs across all N samples; this reflects a ceiling effect where single-sample F1 already exceeds 0.90. WNUT-17 shows 55% identical outputs, driven by the 52% zero-F1 rate.

Oracle-improvable fraction. Across SciERC configurations, 39–42% of instances contain at least one sample with F1 exceeding single-sample[6]: Qwen NER 42.0% (222/529, mean headroom +31.9 pp among improvable instances), Qwen RE 41.3% (164/397, +45.3 pp), LLaMA NER 39.3% (208/529, +33.7 pp). On CoNLL, only 4–9% of instances are improvable, consistent with the ceiling effect.

Diversity-stratified selection gap. On the 44% of Qwen SciERC instances with high diversity ($\geq$4 unique outputs among 8 samples, n=232), oracle headroom reaches +16.3 pp, yet all consistency-based signals (SJ, FK, EM, VC) select at or below single-sample, and LP, the only per-sample signal, achieves merely +0.8 pp. LLaMA replicates this pattern: on its high-diversity subset (n=186), oracle headroom reaches +20.1 pp while consistency signals remain at or below single-sample and LP captures only +2.0 pp. This rules out insufficient diversity as an explanation for the correlation-selection gap.

[6]Oracle-improvable and diversity-stratified analyses use n=529 (excluding instances with empty gold annotations; see §3.3); the diversity quantification in Table 14 uses the full test set.

Table 15: Instance degeneracy statistics (T=1.0). "Constant F1": all N samples receive identical instance-level F1. "Constant utility": MBR (SJ) utility identical across candidates. 4B data confirms degeneracy scales with model capability.

	SciERC 8B (N=8)	SciERC 4B (N=8)	CoNLL 8B (N=16)
Constant F1	17.8%	10.4%	48.4%
Constant utility	13%	7.0%	29%
Oracle gap closed (non-deg.)	23%	25.2%	41%
Within-inst. SJ ρ (median)	0.524	0.550	0.994

Table 16: TOST equivalence testing: signals vs. single-sample on SciERC (N=8, T=1.0). ✓: equivalent at $p<0.05$; ×: not equivalent. Δ: mean F1 difference (signal − single-sample).

	NER				RE			
Signal	Δ	±2	±3	±5	Δ	±2	±3	±5
LP	+0.6	✓	✓	✓	−0.9	×	✓	✓
SJ	+0.1	✓	✓	✓	−1.8	×	×	✓
FK	−0.5	×	✓	✓	−1.6	×	×	✓
EM	−0.8	×	✓	✓	−3.2	×	×	×
VC	−1.0	×	×	✓	−4.5	×	×	×

Instance degeneracy statistics. Table 15 quantifies instance degeneracy across configurations. "Constant F1" denotes instances where all N samples receive identical instance-level F1; "Constant utility" denotes instances where MBR (SJ) utility is identical across candidates, preventing ranking.

TOST equivalence testing. The primary evidence against selection gains comes from effect sizes and confidence intervals: Cohen's d=0.038 for the best signal (NER LP) and bootstrap 95% CI [−3.0, +4.2] pp spanning zero. Table 16 supplements this with two one-sided tests (TOST), testing whether each signal's selection F1 is equivalent to single-sample within a bound δ. The δ values are contextualized against oracle headroom (12–18 pp): δ=5 pp corresponds to forgoing 28–38% of the available headroom, δ=3 pp forgoing 17–23%, and δ=2 pp less than 15%. These proportions use the same test data that defines the headroom, introducing circularity: δ is anchored to the very quantity it contextualizes. We address this below with externally anchored equivalence margins. The breakpoint δ^*, the minimum equivalence margin at which each signal passes TOST, ranges from approximately 1.3 pp (SJ, ∼10% of oracle headroom) to 3.4 pp (FK, ∼26%) on NER, and from approximately 2.9 pp (LP, ∼16%) to 6.5 pp (VC, ∼36%) on RE. On NER (n=529), at δ=±2 pp, SJ and LP are confirmed equivalent to single-sample (TOST $p<0.05$); at δ=±3 pp, four of five signals pass; at δ=±5 pp, all five pass. On RE (n=397, preliminary), EM and VC fail equivalence even at δ=±5 pp due to systematic underperformance (Δ=−3.2 pp and −4.5 pp respectively). The effect sizes are negligible (d=0.038) and confidence intervals span zero, with TOST confirming equivalence at practically meaningful margins.

External equivalence margins. The oracle-headroom anchors above introduce circularity, as a reviewer correctly noted: δ is calibrated against the very quantity it contextualizes. To resolve this, we supplement with three externally grounded margins: (1) δ_{SEM}: the bootstrapped standard error of single-sample F1 (0.43–1.33 pp depending on configuration), which represents measurement noise; (2) $\delta_{\text{practical}}$=0.5 pp, the threshold below which NER differences are considered negligible (Ratinov and Roth, 2009); (3) $\delta_{\text{literature}}$=1.0 pp, the approximate significant-difference threshold established in NER evaluation methodology (Berg-Kirkpatrick et al., 2012).

Results under these external anchors diverge sharply by configuration. On LLaMA CoNLL (n=2,756), all signals pass equivalence at every δ ($p<0.001$), consistent with the near-zero selection deltas ($|\Delta|\leq$0.1 pp) and tight CIs at this sample size. On Qwen CoNLL (n=2,756), no signal passes at any external δ: selection F1 degrades by 0.9–1.5 pp relative to single-sample, which exceeds even $\delta_{\text{literature}}$. On SciERC NER ($n$=529), only SJ passes at δ_{SEM}=1.33 pp (the widest SEM bound); all other signals and tighter margins fail. These external anchors confirm that the equivalence conclusion is configuration-dependent: LLaMA CoNLL selection is robustly equivalent to single-sample under any reasonable margin, while SciERC and Qwen CoNLL selection shows detectable degradation at externally calibrated thresholds.

Multiple comparison considerations. Our primary selection analysis involves ∼20 comparisons (5 signals × 3–4 configurations). Since all primary selection results are non-significant ($p>0.05$), applying Benjamini-Hochberg correction can only make these results more conservative, strengthening the "no signal beats greedy" conclusion. The sole positive finding, the non-tied LP subset (+2.9 pp, p=0.004; §5), survives Bonferroni correction over 5 signal comparisons (p_{adj}=0.020 <

Table 17: Multi-seed quality estimation on SciERC (N=8, T=1.0, seeds: 42, 123, 456). NER: both models; RE: Qwen only (preliminary). Signal ranking is stable across seeds ($\sigma \leq 0.025$). σ is population standard deviation over 3 seeds.

	LP	FK	SJ	VC	EM
NER Spearman ρ (mean $\pm\sigma$)					
Qwen	.217±.013	.254±.016	.364±.018	.386±.019	.307±.008
LLaMA	.162±.025	.256±.024	.325±.021	.384±.014	.279±.011
RE Spearman ρ (mean $\pm\sigma$, Qwen)					
Qwen	.297±.022	.290±.012	.249±.002	.309±.024	.135±.016
NER AUROC (mean $\pm\sigma$)					
Qwen	.579±.008	.600±.016	.659±.016	.677±.011	.631±.013
LLaMA	.565±.008	.593±.006	.632±.012	.673±.009	.606±.005

0.05), as noted in the main text footnote. For quality estimation signal rankings, differences between the top signals (e.g., VC vs. SJ on SciERC) should be interpreted as descriptive rather than as statistically confirmed orderings, given the limited power of 3-seed comparisons.

G Multi-Seed Stability

Table 17 reports quality estimation metrics across three random seeds for both model families on SciERC (N=8, T=1.0). On NER, the signal ranking VC>SJ>EM>FK>LP is identical across all six model-seed combinations, with $\sigma \leq 0.025$ for all signals (rankings are stable across our 3 seeds, though statistical power with n=3 is limited). On RE (Qwen only, preliminary), VC remains the top signal (ρ=0.309±0.024), but LP rises to #2 (ρ=0.297±0.022), above FK, SJ, and EM. This reversal from the NER ranking reflects RE's higher sensitivity to per-sample scoring. LLaMA absolute values are systematically lower than Qwen on NER, consistent with the directionally-consistent-but-magnitude-dependent characterization in the main text.

H Training Configuration

Both models are fine-tuned with identical hyperparameters (Table 18) to ensure comparisons across model families reflect model differences rather than training confounds.

Prompt template. For NER, we use a structured prompt:

> `System`: You are an expert in Named Entity Recognition.
> `User`: Extract all named entities from the following text. Output as JSON.
> Text: `{input_text}`
> Entity types: `{schema_types}`

RE follows the same structure with "Relation Extraction" and relation type lists substituted accordingly.

Table 18: Training configuration. Both Qwen3-8B and LLaMA-3.1-8B-Instruct use identical settings unless noted.

Parameter	Value
LoRA rank / α	32 / 64
LoRA target	All linear modules
LoRA dropout	0
Optimizer	AdamW
Learning rate	2e-4, cosine schedule
Warmup ratio	0.1
Effective batch size	16 (2 × grad. accum. 8)
Epochs	5
Max sequence length	2048
Precision	bf16
Gradient checkpointing	Enabled
Max gradient norm	1.0
Training seed	42
Framework	LLaMA Factory
Dataset-specific	
SciERC split	1861 / 275 / 551
CoNLL / WNUT	+ load_best_model_at_end, eval_steps=200, save_limit=5

Constrained decoding. Inference uses vLLM with XGrammar-based JSON schema enforcement (Dong et al., 2024; Willard and Louf, 2023), which constrains outputs to valid entity/relation structures matching the target schema. This eliminates format errors but may compress per-sample diversity by restricting the output space.[7]

I Thinking Token Impact Analysis

Qwen3-8B's SciERC checkpoint was trained with empty `<think></think>` blocks prepended to each response (§4.1). We quantify the potential impact on LP scores through three complementary arguments.

LP shift magnitude. Empty thinking tokens (mean 4.5 per sample) constitute 2.1% of tokens on SciERC. The estimated LP shift is 0.001 nats per token (2.68% of the median within-instance LP range of 0.041 nats). Because all samples share

[7] SciERC training used `enable_thinking=True` (LLaMA Factory 0.9.4 default), injecting empty `<think></think>` blocks (4–5 tokens) into each training response with loss computation; CoNLL and WNUT explicitly set `enable_thinking=False`. Constrained decoding at inference time enforces JSON-only output, suppressing thinking tokens. See §4.1 and Limitations for discussion of this confound.

identical thinking-token LP offsets, cumulative LP rankings are invariant; mean LP changes selection for $<1.5\%$ of instances.

Constant prefix cancellation. All N samples within an instance share the same prompt and, under constrained decoding, the same absence of thinking tokens at inference time. Any LP bias from the training-time thinking prefix is therefore a near-constant additive shift across all candidates within an instance, leaving within-instance LP *rankings* unchanged.

CoNLL control group. CoNLL and WNUT-17 checkpoints were trained without thinking tokens (`enable_thinking=False`). If thinking tokens caused LP compression on SciERC, CoNLL should exhibit lower tied fractions. Instead, CoNLL shows *higher* tied fractions at both temperatures (86.8% at T=1.0, 95.8% at T=0.7) than SciERC (60.3% at T=1.0), directly contradicting the thinking-token hypothesis.

J Log-Probability Score Compression

Under constrained decoding, log-probability scores are severely compressed within instances, limiting per-sample discrimination (Table 19, Figure 8). The median within-instance LP range is only 0.041 nats; we use ε=0.05 as an illustrative threshold to characterize this compression. At this cutpoint, 60.3% of instances (n=529, excluding empty gold annotations) have a within-sample range below ε, and 75% of pairwise comparisons are effective ties ($|\Delta|<0.026$). The CDF rises smoothly with no cliff or discontinuity (Figure 8); compression is a pervasive regime rather than a bimodal split. This compression is stable across seeds: seed=456 yields median LP range=0.040 nats and tied fraction=62.2% (vs. seed=42: 0.041 nats, 60.3%). Seed 456 uses a different subtask flag (joint vs. ner for seeds 42 and 123); the LP distributions are statistically indistinguishable (median LP: 8.37 vs. 8.20 nats; Mann–Whitney p=0.55); the subtask change does not affect compression. For comparison, SJ has only 12.5% tied instances at the same threshold; SJ's advantage over LP for selection is one of score resolution, not information content.

Non-tied subset. Among instances with LP range $\geq$0.05 (n=210, 40% of total), log-probability selection *significantly* outperforms single-sample (+2.9 pp F1, p=0.004), and

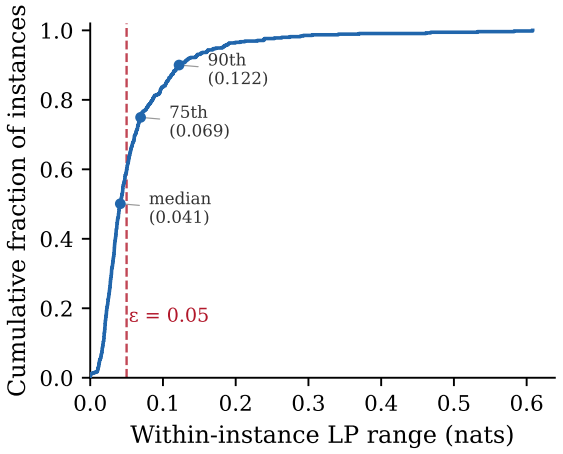


Figure 8: CDF of within-instance LP range on SciERC NER (N=8, n=529). The median LP range is 0.041 nats (dashed line), below ε=0.05. The smooth CDF confirms compression is pervasive, not bimodal.

Table 19: Log-probability score compression on SciERC NER (N=8, n=529). "Tied" = fraction of instances where within-instance LP range $<\varepsilon$.

ε threshold	Tied instances (%)
0.01	1.1%
0.05	60.3%
0.1	∼84%
0.5	∼99%

within-instance $\rho(\text{LP}, \text{F1})$ rises from 0.08 (tied subset) to 0.39 (non-tied subset). As a continuous complement to the binary tied/non-tied analysis, pooled within-instance $\rho(\text{LP}, \text{F1})$=0.248 across all instances confirms weak but positive per-sample signal; however, only 21.6% of instances achieve $p<0.05$ significance, and $\rho(\text{LP range}, \text{F1 range})$=0.328 shows that instances with more LP spread also exhibit more F1 variability. LP carries information, but compression limits its exploitation. A free-form NER decoding ablation (Appendix P) confirms this compression persists without constrained decoding.

ε sensitivity sweep (post-hoc exploratory). Table 20 sweeps the tied/non-tied threshold from ε=0.01 to 0.20 as a post-hoc exploratory analysis characterizing how selection gain varies with compression severity. The tied fraction increases smoothly with no cliff or discontinuity, and the selection gain Δ grows monotonically as the non-tied subset narrows to progressively higher-spread instances. ε=0.05 yields the strongest statistical separation (p=0.004); at $\varepsilon\geq$0.15, the non-tied subset

Table 20: ε sensitivity sweep for LP selection on SciERC NER (N=8, seed=42, n=529). "Tied%" = fraction with within-instance LP range $<\varepsilon$. Δ: selection F1 gain over single-sample on the non-tied subset. The tied fraction increases smoothly; ε=0.05 yields the strongest statistical separation.

ε	Tied%	Non-tied n	Δ (pp)	p
0.01	1.1%	523	+0.58	.21
0.02	13.6%	457	+0.83	.15
0.03	31.8%	361	+1.31	.079
0.05	**60.3%**	**210**	**+2.86**	**.004**
0.10	83.9%	85	+3.69	.034
0.15	93.2%	36	+4.15	.11
0.20	96.4%	19	+4.59	.13

Table 21: Maximum selection F1 variation (pp) across $\varepsilon \in [0, 0.2]$. At ε=0.05 (used throughout), consistency signals change by $\leq$1.09 pp; EM is invariant (integer scoring). LP is most sensitive as ε governs its tied/non-tied partition. SciERC: n=529; CoNLL: n=2,756.

	SciERC		**CoNLL**	
Signal	Qwen	LLaMA	Qwen	LLaMA
SJ	0.92	1.45	0.99	0.18
FK	1.03	0.93	0.60	0.15
VC	2.33	0.53	0.72	0.18
EM	0.00	0.00	0.00	0.00
LP	3.65	2.58	3.00	0.31

becomes too small for reliable inference ($n \leq 36$).

ε robustness of selection F1. The preceding analysis stratifies LP by a tied/non-tied threshold; consistency-based signals (SJ, FK, VC, EM) also use ε to define MBR-style proximity in selection. Table 21 reports the maximum selection F1 variation across $\varepsilon \in [0, 0.2]$ for all signals and configurations. At ε=0.05, consistency signals change by at most 1.09 pp (mean 0.21 pp), with EM completely ε-invariant (integer scoring); LP is the most sensitive (up to 3.65 pp across the full range), as expected. Spearman ρ, the primary quality estimation metric throughout, is a continuous rank correlation that does not depend on any discretization threshold; the ε parameter affects only the selection analysis.

Non-tied subset analysis. Table 22 presents the non-tied subset analysis referenced in §5.

Within-instance LP-quality alignment. Table 23 reports median within-instance $\rho(\text{LP}, \text{F1})$ across datasets, computed over non-degenerate instances (those with $\geq$2 distinct F1 values across N=8 samples). Few-NERD exhibits substantially stronger alignment (ρ=0.667) than SciERC (ρ=0.262), directly explaining why LP selection succeeds only on Few-NERD. CoNLL alignment is moderate ($\rho \approx 0.51$), but the valid-instance fraction is low (41% for Qwen, 11% for LLaMA) because degeneracy eliminates most headroom before alignment can operate. LP-quality alignment is the primary predictor of selection direction; degeneracy constrains headroom but does not independently predict whether LP selection gains materialize (§5).

Table 22: Non-tied subset analysis (SciERC, Qwen3-8B, N=8, T=1.0). Instances split at LP range ε=0.05. Within-instance ρ and selection gain both concentrate in the non-tied subset. Full ε sweep in Table 20.

Subset	n	ρ(LP,F1)	LP Δ vs. greedy
Tied (range $<$0.05)	319	0.08	$\approx$0
Non-tied (range $\geq$0.05)	210	0.39	+2.9 pp (p=0.004)
All instances	529	0.248	+0.2 pp (p=0.29)

Table 23: Within-instance LP-quality alignment across datasets (N=8, T=1.0). ρ: median Spearman correlation between LP and F1 within each instance, over non-degenerate instances (n_{valid}: $\geq$2 distinct F1 values).

Dataset	Model	Med. ρ	n_{valid} / n_{total}	Valid%
SciERC	Qwen3-8B	0.262	435 / 529	82%
Few-NERD	Qwen3-8B	0.667	28,055 / 32,565	86%
CoNLL	Qwen3-8B	0.506	1,124 / 2,756	41%
CoNLL	LLaMA-3.1-8B	0.512	298 / 2,756	11%

K Consistency-Stratified Selection Analysis

Table 24 stratifies SciERC instances by voting confidence (VC) quartiles and reports selection Δ (gain over single-sample) for LP and SJ.

The NER pattern reveals two complementary failure modes rather than a monotonic relationship between consistency and selection gain:

- **Q4 ceiling effect.** High-consistency instances (mean VC=0.92) have oracle headroom of only 4.78 pp. Samples have already converged, leaving little room for any signal to improve selection.
- **Q1–Q2 noise.** Low-consistency instances retain large headroom (14–17 pp) but LP rankings are unreliable, and LP rankings produce inconsistent Δ signs across quartiles.

On †RE, the pattern differs: Q4 LP achieves +4.01 pp; LP may recover selection ability when

Table 24: Selection Δ (pp over single-sample) stratified by VC quartile on SciERC (N=8, Qwen3-8B). Oracle headroom collapses at Q4; LP rankings are unreliable at Q1–Q2. Neither quartile supports consistent selection gains.

Quartile	Mean VC	LP Δ	SJ Δ	Oracle (pp)
NER				
Q1 (low)	0.30	+1.55	+0.54	16.85
Q2	0.51	−0.65	−1.89	14.35
Q3	0.68	+1.97	+1.84	16.83
Q4 (high)	0.92	−0.41	−0.04	4.78
†*RE*				
Q1 (low)	—	−3.56	—	—
Q4 (high)	—	+4.01	—	—

high consistency coincides with structured output diversity. This is a single-dataset observation and should be interpreted cautiously.

On CoNLL NER, Q3–Q4 instances have VC=1.0 (all samples identical), and Δ=0 universally; this ceiling effect is even more extreme than SciERC Q4.

L Exploratory Extension to Relation Extraction

We extend the NER evaluation to relation extraction (RE) on SciERC as an exploratory single-benchmark analysis. RE produces sets of (head, relation, tail) triples with partial overlap; 40% of RE instances receive zero F1, and entity span errors account for 55.5% of RE false negatives, a prerequisite failure that consistency signals cannot detect (§D). All RE results use Qwen3-8B unless noted; single-seed (seed=42) for N=8 analyses, 3-seed for N=16.

Quality estimation. RE quality estimation is reported in Table 17 (3-seed means). The signal ranking differs from NER: VC leads (ρ=0.309±0.024), but LP rises to #2 (ρ=0.297±0.022), above FK, SJ, and EM, a reversal reflecting RE's higher sensitivity to per-sample scoring. RE miscalibration is more severe than NER (ECE 0.109–0.339; Figure 5).

Selection. Table 25 presents RE selection at N=8: no signal outperforms single-sample (0.404) on the full set (n=397) or the non-zero subset (n=238), with 18.1 pp of unrealized oracle headroom. At N=16 (Qwen3-4B, Table 26), LP is the sole positive signal (Δ=+0.7 pp, p=0.058), consistent with LP being the strongest RE selection signal across scales.

Table 25: RE selection F1 (macro, instance-averaged) on SciERC (N=8, T=1.0). Full set (n=397) and non-zero F1 subset (n=238). No signal outperforms greedy on either subset.

Method	Full (n=397)	Non-zero (n=238)
Oracle	0.585	0.757
LP	0.395	0.600
SJ	0.386	0.604
EM	0.372	0.583
VC	0.359	0.574
Greedy	0.404	0.674

Scale ablation. At 4B scale (Qwen3-4B, N=8, seed=42), RE quality estimation shows FK ρ=0.359 and LP ρ=0.351, both above their 8B counterparts (FK 0.275, LP 0.266), while SJ declines (0.267 vs. 0.250). Selection remains below single-sample on RE (Δ up to −3.4 pp) despite 17.1 pp of oracle headroom.

Free-form decoding. RE LP selection consistently reverses direction under free-form decoding (+1.8±0.1 pp across 4 seeds vs. −0.9 pp under constrained decoding at 8B; §P); constrained decoding's fixed template likely obscures LP signal in RE (§O).

M RE Conditional Selection

Table 25 extends the selection analysis to the RE non-zero F1 subset. Of n=397 RE instances, 238 (60%) have non-zero F1 and 159 (40%) have zero F1. All five signals underperform single-sample on both subsets. On the non-zero subset, selection performance is *worse* than on the full set (e.g., VC: −10.0 pp on non-zero vs. −4.5 pp on full set), because single-sample is already strong on non-zero instances (F1=0.674) and signals actively select inferior samples. Oracle headroom exists (0.757 vs. 0.674) but no signal exploits it, ruling out the 40% zero-F1 floor effect as the explanation for RE selection failure.

N RE N=16 Selection Analysis

Table 26 extends the selection analysis to RE at N=16 with 3-seed statistics. All four consistency signals underperform single-sample, with EM showing the largest deficit (Δ=−1.5 pp, Fisher p>0.99). LP is the only positive signal (Δ=+0.7 pp, Fisher p=0.058), driven primarily by seed 456 (Δ=+1.5 pp, p=0.026); the remaining seeds show smaller effects (+0.1 and +0.6 pp,

Table 26: RE best-of-N selection F1 (macro, instance-averaged) on SciERC (Qwen3-4B, N=16, T=1.0, 3-seed mean±σ, n=551). Fisher combined p vs. greedy. LP is the only signal with positive Δ. σ is population standard deviation over 3 seeds.

Method	F1	Δ (pp)	p(Fisher)
Oracle	.463±.006	+17.1	—
LP	**.299**±.007	+0.7	.058
VC	.284±.002	−0.8	.973
FK	.283±.005	−0.9	.985
SJ	.281±.005	−1.0	.993
EM	.277±.007	−1.5	.999
Greedy	.291±.000	—	—
Random	.262±.001	−2.9	—

p>0.20). Even the most favorable seed captures only ∼9% of the 17.1 pp oracle headroom. This cross-task consistency (LP as the only viable selection signal on both NER and RE) reinforces the persistent limitation of instance-level signals for candidate ranking across tested regimes.

O MBR Decoding and Task-Specific Interactions

Connection to MBR decoding. Our selection strategies instantiate minimum Bayes risk (MBR) decoding (Eikema and Aziz, 2022; Müller and Sennrich, 2021): each selects $\hat{y} = \arg\max_{y_i} \sum_{j \neq i} U(y_i, y_j)$ with utility U defined by SJ, EM, or entity overlap. Despite MBR's success in machine translation (Freitag et al., 2022) and its extension to summarization, code generation, and captioning (Jinnai et al., 2024), all variants fail here: despite positive within-instance utility–F1 correlation, selection gains are limited by instance degeneracy, the core of the correlation-selection gap. Eikema et al. (2025) study structure-conditional MBR decoding and find that standard utility functions degrade on structured outputs, consistent with our finding that consensus-proximity selection fails in IE despite its success in machine translation.

Task-specific decoding interactions. The free-form ablation reveals divergent behavior across tasks. On RE, LP selection flips from −0.68 pp (constrained) to +1.69 pp (free-form; preliminary, single RE benchmark), a reversal that spans all 7 relation types and is driven by selective variance reallocation (p<0.0001). The win/loss ratio shifts from 36:59 (constrained) to 45:45 (free-form), with the gains concentrated on relation-dense instances (mean 3.3 gold relations). This interaction does not generalize to NER, where selection remains negligible under both regimes. This ablation uses a single seed (seed=42); whether the RE reversal replicates across seeds and generalizes to other RE benchmarks warrants further investigation. The non-tied subset (n=210) is partially reshuffled under free-form decoding: Jaccard similarity between the constrained and free-form non-tied sets is 0.472: 87 instances newly gain LP discrimination while others lose it, and the newly non-tied subset shows a 3× increase in within-instance LP range for the newly non-tied subset. Despite this reshuffling, aggregate NER selection gains remain negligible. Score compression is not a fixed property of specific instances but rather a pervasive regime that shifts across instances without producing net selection improvement. Free-form decoding also slightly improves quality estimation (NER VC ρ: 0.379→0.409), though this does not translate to selection gains.

P Free-Form Decoding Ablation

Table 27 compares constrained and free-form decoding across four dimensions on SciERC (N=8, T=1.0, Qwen3-8B; seed=42 shown, 4-seed RE statistics below). Free-form decoding removes grammar constraints entirely; outputs are parsed post-hoc (parse rate 99.95%, 4406/4408). LP score distributions are nearly identical across regimes (Panel A: within-instance std ratio 0.97×), and NER selection gains remain negligible under both regimes (Panel B). RE LP selection consistently reverses direction under free-form decoding across 4 seeds: +1.84±0.12 pp (seeds 42/123/456/789: +1.69, +1.76, +2.00, +1.91 pp) vs. −0.68 pp constrained. The constrained reference (−0.68 pp) uses the ablation-matched baseline; the primary experiment (Table 2) yields −0.9 pp on the same gold-filtered set. Quality estimation improves modestly for most signals under free-form decoding (Panel C), with the largest gain for FK (+0.053 NER, +0.039 RE).

CoNLL NER free-form results. Extending the free-form ablation to CoNLL NER (Qwen3-8B, 4 seeds, N=8, T=1.0), LP range median is 0.058±0.007 across seeds, higher than SciERC (0.041) but still compressed given CoNLL's 86.8% tied fraction under constrained decoding. Free-form decoding reduces the tied fraction modestly, yet selection remains below single-sample:

Table 27: Comparison of constrained vs. free-form decoding on SciERC (Qwen3-8B, N=8, T=1.0, seed=42; RE selection gap replicates across 4 seeds: $+1.84 \pm 0.12$ pp).

PANEL A: LOGPROB DISTRIBUTION

Metric	Constrained	Free-form	Δ
Mean LP std	0.0206	0.0199	−0.0007
Mean LP range	0.0623	0.0599	−0.0024
Tied fraction (ε=0.05)	59.7%	57.7%	−2.0 pp

PANEL B: SELECTION F1

Task	Method	Constrained	Free-form	Δ
NER	Greedy	0.6182	0.6204	+0.0022
	LP-selected	0.6241	0.6222	−0.0019
	Selection gap	+0.0059	+0.0018	−0.0041
RE	Greedy	0.2913	0.2880	−0.0033
	LP-selected	0.2845	0.3050	+**0.0205**
	Selection gap	−0.0068	+0.0169	+**0.0237**

PANEL C: QE SIGNAL QUALITY (SPEARMAN ρ)

	NER (n=529)			RE (n=397)		
Signal	Constr.	Free	Δ	Constr.	Free	Δ
SJ	0.360	0.377	+0.018	0.250	0.273	+0.023
FK	0.267	0.320	+0.053	0.275	0.315	+0.039
LP	0.205	0.213	+0.008	0.266	0.304	+0.038
EM	0.295	0.317	+0.023	0.134	0.137	+0.002
VC	0.379	0.409	+0.030	0.350	0.323	−0.026

PANEL D: PARSE RATE (FREE-FORM)

Metric	Value
Total samples	4408
Parse rate	99.95%
Instances 8/8 valid	549 / 551 (99.6%)

LP-selected F1=0.896 vs. single-sample 0.904 (Δ=−0.77 pp). Quality estimation under free-form decoding yields SJ ρ=0.461, EM ρ=0.471, LP ρ=0.326, comparable to constrained results (Table 1). The failure to improve selection despite reduced compression is consistent with CoNLL's low oracle headroom (5.0 pp at N=8), which limits the selection space regardless of score distribution quality.

Q LP Selection Failure Stability

We analyze RE LP selection failures across three random seeds (constrained decoding, N=8, T=1.0, Qwen3-8B) to assess whether failures are systematic or stochastic.

Pairwise overlap. LP selection failures (instances where LP-selected F1 < single-sample F1) show moderate cross-seed stability: the average pairwise Jaccard similarity is 0.42 (range: 0.38–0.46); failures are partially systematic (driven by instance complexity) but not fully deterministic. The 3-way intersection contains 34 instances (27.9% of any-seed failures), which constitutes a stable failure core.

Instance complexity drives failure. Table 28 stratifies instances by failure stability. Stable failures (all 3 seeds) involve more gold relations (2.12 vs. 1.62 for never-fail instances) and higher single-sample F1 (0.688 vs. 0.429); LP mis-selects on instances where the model already performs well,

Table 28: RE LP selection failure stability across 3 seeds (SciERC, N=8). Stable failures concentrate on relation-dense instances with high single-sample F1.

Group	n	Gold rels	Greedy F1	LP ΔF1
Stable fail (3 seeds)	34	2.12	0.688	−0.499
Fail 2+ seeds	26	2.46	0.774	−0.450
Fail 1 seed	62	2.32	0.692	−0.239
Never fail	429	1.62	0.429	+0.041

Table 29: Effect of sample pool size on signal quality (Qwen3-8B, SciERC, 3-seed mean Spearman ρ).

	NER ρ			**RE** ρ		
Signal	N=8	N=16	Δ	N=8	N=16	Δ
SJ	0.3641	0.4105	**+0.0464**	0.2485	0.2758	**+0.0273**
FK	0.2536	0.2833	**+0.0297**	0.2903	0.2964	**+0.0061**
VC	0.3861	0.4092	**+0.0231**	0.3088	0.3738	**+0.0650**
LP	0.2167	0.2393	**+0.0226**	0.2970	0.3100	**+0.0130**
EM	0.3071	0.3395	**+0.0324**	0.1353	0.1506	**+0.0153**
Mean	0.3055	0.3364	**+0.0308**	0.2560	0.2813	**+0.0253**

Table 30: Cross-seed signal ranking stability (Kendall's W, 3 seeds).

Model	Task	N	Kendall's W	Consensus ranking
Qwen	NER	8	1.000	VC > SJ > EM > FK > LP
Qwen	NER	16	0.956	SJ ≈ VC > EM > FK > LP
Qwen	RE	8	0.822	VC > LP > FK > SJ > EM
Qwen	RE	16	0.956	VC > LP > FK > SJ > EM
LLaMA	NER	8	1.000	VC > SJ > EM > FK > LP

Table 31: Cross-seed standard deviation (σ) of Spearman ρ at $N = 8$ vs. $N = 16$ (Qwen3-8B, SciERC, 3 seeds). σ is population standard deviation.

	NER σ_ρ		**RE** σ_ρ	
Signal	N=8	N=16	N=8	N=16
SJ	0.0179	0.0102	0.0015	0.0067
FK	0.0157	0.0079	0.0121	0.0143
VC	0.0187	0.0260	0.0244	0.0158
EM	0.0076	0.0031	0.0157	0.0049
LP	0.0125	0.0017	0.0218	0.0152
Mean	0.0145	0.0098	0.0151	0.0114

precisely the instances where selection would be most valuable. Failure rate increases monotonically with gold relation count: 4.3% at 1 relation, 20.7% at 3 relations, 26.2% at 6 relations.

Flip instances. Only 5 RE instances with gold relations flip from LP failure (constrained) to LP success (free-form), with mean gold relation count 3.67 (well above the population mean of 1.86). These instances show large constrained LP errors (Δ=−0.430) that reverse under free-form decoding (Δ=+0.503), but their rarity (5/397) confirms that the RE selection flip is driven by aggregate variance reallocation rather than systematic instance-level correction.

R N-Scaling Detailed Analysis

Tables 29–31 provide detailed N-scaling analysis on Qwen3-8B SciERC with 3-seed statistics.

Signal rankings are consistent across our 3 seeds (Kendall's W=1.000 for NER at N=8, both models), dropping to W=0.956 at N=16 where SJ and VC converge (Table 30); however, with only n=3 seeds statistical power is insufficient to confirm stability, and additional seeds may reveal ranking changes. Doubling N reduces cross-seed σ by 32% on NER and 24% on RE (Table 31), with the largest reductions for SJ (−43% NER) and EM (−59% NER); additional samples stabilize structural signals more than LP.

N-scaling: selection headroom vs. selection gain. Table 32 traces the effect of sample count on degeneracy, oracle headroom, and per-signal selection gain. Oracle headroom grows roughly linearly with N (+4.4 pp at N=2 to +20.0 pp at N=32). LP selection improves monotonically with N (+0.39 pp at N=8 to +0.95 pp at N=32), but even at N=32, where degeneracy falls to 4% and oracle headroom reaches ∼20 pp, the best signal captures less than 5% of the available improvement. Degeneracy drops monotonically (43% at N=2 to 4% at N=32), yet reduced degeneracy does not unlock selection, consistent with the LoRA-rank ablation finding (§4.5). VC selection degrades with increasing N (−2.66 pp at N=2 to −4.35 pp at N=32), consistent with mode-seeking bias in voting-based selection as the candidate pool grows.

Bootstrap significance at N=32. Table 33 reports bootstrap confidence intervals and p-values for selection at N=32 (B=10,000, n=529). No signal achieves significant improvement over greedy at α=0.05 (LP: p=0.091; FK: p=0.101), while VC is significantly *worse* (−4.36 pp, p=1.0). However, all signals except VC significantly outperform random selection (p<0.001, permutation test); the signals carry meaningful quality informa-

[8] N-scaling analysis uses gold-filtered evaluation (n=529, greedy F1=0.640). Values differ slightly from Table 2 (0.645) and Table 3 (0.644) due to seed and aggregation differences; relative trends are consistent.

Table 32: Effect of sample count on selection (SciERC NER, Qwen3-8B, T=1.0, gold-filtered, seed=42; exp_025).[8] "Best Δ": highest selection gain across all 5 signals. Per-signal Δ in pp vs. greedy.

N	Degen.%	Oracle (pp)	LP	SJ	FK	EM	VC	Best Δ
2	42.7%	+4.4	−0.89	−2.72	−2.72	−2.72	−2.66	−0.89
4	21.0%	+10.2	+0.03	−0.31	−0.40	−0.74	−2.16	+0.03
8	11.0%	+14.5	+0.39	+0.20	+0.15	−0.77	−2.87	+0.39
16	7.4%	+17.7	+0.78	+0.24	−0.02	−0.06	−3.95	+0.78
32	4.0%	+20.0	+0.95	+0.35	+0.65	+0.22	−4.35	+0.95

Table 33: Bootstrap significance test at N=32 (SciERC NER, Qwen3-8B, T=1.0, B=10,000, n=529 gold-non-empty instances; 22 instances with no gold entities are excluded as they contribute zero selection differential). Δ: selection gain vs. greedy. CI: 95% bootstrap percentile interval. p_{greedy}: one-sided p-value vs. greedy; p_{rand}: permutation test vs. random selection.

Signal	Δ (pp)	95% CI	p_{greedy}	p_{rand}
LP	+0.95	[−0.41, +2.34]	0.091	<0.001
FK	+0.66	[−0.36, +1.67]	0.101	<0.001
SJ	+0.36	[−0.66, +1.38]	0.244	<0.001
EM	+0.22	[−0.99, +1.44]	0.355	<0.001
VC	−4.36	[−6.26, −2.37]	1.000	0.741
Oracle	+20.00	[+17.98, +22.10]	<0.001	—

tion yet fail to translate it into reliable selection gains. The oracle gap (+20.0 pp, p<0.001) confirms substantial unrealized headroom.

S Temperature Ablation

Table 34 reports the effect of sampling temperature on QE signals, oracle headroom, and output quality across four temperatures on SciERC NER.

Oracle headroom increases monotonically with temperature (+9.1 pp at T=0.5 to +14.6 pp at T=1.2), confirming that higher temperature consistently expands the oracle ceiling under constrained structured decoding. Structural QE signals improve with temperature: SJ ρ rises from 0.297 (T=0.5) to 0.377 (T=1.2), consistent with more diverse outputs providing richer pairwise evidence. LP ρ peaks at T=0.8 (0.253) then degrades to 0.196 at T=1.2. This divergence confirms that LP compression is model-intrinsic: higher temperature increases between-instance LP variance (benefiting structural signals) but does not resolve within-instance compression, degrading LP's discriminative power. Inter-sample agreement (κ) decreases monotonically (0.577 to 0.356); higher temperature generates more diverse outputs. Mean sample F1 also decreases (0.636 to 0.594), consistent with the quality–diversity trade-off. The T=0.7 and T=1.0 comparison in Table 8 is consistent with the trends reported here.

Table 34: Temperature ablation on SciERC NER (Qwen3-8B, N=8, seed=42). Oracle gap and QE ρ across four temperatures. κ: Fleiss' κ (inter-sample agreement). T=0.5 and T=1.0 use joint (NER+RE) subtask; T=0.8 and T=1.2 use NER-only.

T	Oracle gap (pp)	SJ ρ	LP ρ	κ	Sample F1
0.5	+9.1	0.297	0.218	0.577	0.636
0.8	+12.5	0.359	0.253	0.451	0.622
1.0	+13.0	0.360	0.205	≈0.39	≈0.612
1.2	+14.6	0.377	0.196	0.356	0.594

Temperature controls degeneracy and LP spread. Table 35 reports instance degeneracy (constant F1, gold-filtered) and median within-instance LP range across four temperatures on SciERC. Both quantities are strictly monotonic in T: degeneracy falls from 41.4% (T=0.5) to 14.9% (T=1.2), while LP range increases from 0.015 to 0.058 nats. Temperature thus provides a direct lever for practitioners to shift the degeneracy regime. However, even at T=1.2 where degeneracy (14.9%) approaches Few-NERD levels (∼14%), within-instance $\rho(\text{LP}, \text{F1})$ remains at 0.225, far below Few-NERD's ∼0.5. This confirms that temperature controls degeneracy but cannot improve LP-quality alignment, which is largely determined by task structure and remains relatively stable across generation configurations (§Z). LP-quality alignment, not degeneracy, is the primary predictor of LP selection direction (§5).

Table 35: Temperature sensitivity of degeneracy and LP spread (SciERC NER, Qwen3-8B, N=8). Degeneracy: constant-F1, gold-filtered. LP range: median within-instance range.

T	Degen.%	LP range (nats)	Fleiss' κ	SJ mean
0.5	41.4	0.015	0.572	0.830
0.8	23.8	0.031	0.471	0.752
1.0	17.8	0.042	0.405	0.700
1.2	14.9	0.058	0.352	0.647

T Cross-Dataset OOD Analysis

To test whether confidence signals generalize under domain shift, we evaluate models on out-of-distribution data: a SciERC-trained model applied to CoNLL instances (wider→narrower schema) and vice versa (Table 36).

Table 36: Cross-dataset OOD quality estimation (Spearman ρ, Qwen3-8B, N=8, T=1.0, seed=42). SciERC→CoNLL: SciERC-trained model on CoNLL instances (n=2756). CoNLL→SciERC: CoNLL-trained model on SciERC instances (n=529, excluded from conclusions due to catastrophic model failure). In-domain ρ from Table 1 (same model, same seed). $^{***}p<0.001$, $^{*}p<0.05$.

	SciERC→CoNLL			CoNLL→SciERC		
Signal	ρ	In-dom.	Δ	ρ	In-dom.	Δ
VC	.469***	.379	+.090	+.105*	.379	−.274
SJ	.378***	.360	+.018	−.186***	.360	−.546
EM	.300***	.295	+.005	−.202***	.295	−.497
FK	.285***	.267	+.018	−.070	.267	−.337
LP	.246***	.205	+.041	−.152***	.205	−.355

Table 37: Verbalized confidence (VerbConf) vs. log-probability selection (Qwen3-8B, N=8, T=1.0, seed=42). Parse fail: fraction of samples where the confidence score could not be extracted.

	SciERC NER		CoNLL NER	
Metric	VerbConf	LP	VerbConf	LP
ρ	0.297	0.205	0.337	0.316
AUROC	0.681	0.578	0.747	0.753
Selection F1	0.620	0.647	0.889	0.904
Δ vs. greedy	−2.45 pp	+0.2 pp	−1.93 pp	−0.77 pp
Mean score	8.58 / 10	—	9.24 / 10	—
Parse fail rate	0.26%	—	0%	—

Wider-to-narrower schema transfer. When the SciERC-trained model (6 NER types) is applied to CoNLL (4 types), all signals remain significant ($p<0.001$) and structural signals improve over their in-domain levels (same model, seed=42, N=8; in-domain ρ from Table 1): VC reaches ρ=0.469 (in-domain 0.379, Δ=+0.090), SJ ρ=0.378 (in-domain 0.360, Δ=+0.018), and FK ρ=0.285 (in-domain 0.267, Δ=+0.018). The wider training schema subsumes CoNLL's narrower schema, so the model produces reasonable outputs even out-of-domain (single-sample F1=0.522), and structural signals benefit from the simpler target schema's reduced output ambiguity.

Narrower-to-wider schema transfer (excluded). The reverse direction is excluded from our conclusions due to catastrophic model failure, not signal failure. The CoNLL-trained model recognizes only 4 news entity types (PER, ORG, LOC, MISC) and has no relation extraction capability; when applied to SciERC's 6 scientific entity types and 7 relation types, the model cannot produce meaningful structured outputs (single-sample NER F1=0.009, RE F1=0.001). Under this regime, most signals flip to negative or near-zero correlations (Table 36); only VC retains weak positive correlation (ρ=0.105, p=0.016). These negative correlations reflect the model systematically generating confident but incorrect outputs, a capability failure analogous to WNUT-17 (§4.2) and not evidence against signal utility.

Asymmetry interpretation. The directional asymmetry is driven by schema coverage: wider→narrower preserves signal utility because the SciERC model's 6-type schema subsumes CoNLL's 4 types, while narrower→wider fails because the CoNLL model lacks the entity and relation types required by SciERC. This extends the capability boundary identified on WNUT-17 to the domain-shift setting: confidence signals require sufficient model capability as a prerequisite, and results from the CoNLL→SciERC direction should not be interpreted as evidence of signal degradation under domain shift.

U Verbalized Confidence

We evaluate verbalized confidence (VerbConf) as a second per-sample signal following Lin et al. (2022) and Tian et al. (2023): after generating each structured output, the model is prompted to rate its confidence on a 1–10 scale. Table 37 reports quality estimation and selection results.

Severe overconfidence. VerbConf exhibits extreme score compression: mean confidence is 8.58/10 on SciERC (single-sample F1=0.645) and 9.24/10 on CoNLL (single-sample F1=0.904), with most scores concentrated in the 8–10 range. This compression is more severe than LP (which spans a 0.041-nat median range); score compression is a general property of model self-assessment, not an artifact of log-probability computation.

QE–selection decoupling. VerbConf achieves higher ρ than LP on both datasets (SciERC: 0.297 vs. 0.205; CoNLL: 0.337 vs. 0.316) and competitive AUROC (SciERC: 0.681 vs. 0.578; CoNLL: 0.747 vs. 0.753), yet yields the worst selection among all signals (Δ=−2.45 pp on SciERC, −1.93 pp on CoNLL). This dissociation further demonstrates the correlation-selection gap: a signal can rank instances by expected quality without being able to discriminate candidates within an in-

Table 38: LLaMA-3.2-3B SciERC NER quality estimation and selection (N=8, T=1.0, seed=42). Greedy F1=0.627, Oracle F1=0.763, headroom=13.63 pp.

Signal	ρ	AUROC	Sel. F1	Δ vs. greedy
VC	**.460**	.743	0.611	-1.54 pp
SJ	.375	.738	**0.633**	$+0.61$ pp
EM	.334	**.776**	—	—
FK	.268	.671	0.630	$+0.28$ pp
LP	.266	.633	0.628	$+0.12$ pp
Greedy	—	—	0.627	—
Oracle	—	—	0.763	—

Table 39: Few-NERD INTRA coarse: complete quality estimation and selection results (Qwen3-8B, N=8, T=1.0, gold-filtered n=32,565). ρ: Spearman correlation with instance F1. Sample AUROC: per-sample LP discrimination. Sel. Δ: selection F1 minus greedy (0.748). CI: bootstrap 95% confidence interval.

Signal	ρ	Sample AUROC	Sel. Δ	95% CI
LP	.220	.578	$+\mathbf{1.39}$ pp	[+1.18, +1.53]
VC	.386	—	-1.79 pp	—
FK	.313	—	-2.26 pp	—
SJ	.300	—	-2.14 pp	—
EM	.287	—	-3.85 pp	—
Greedy: 0.748		Oracle: 0.872	Headroom: 12.4 pp	

stance. The AUROC–ρ decoupling on CoNLL (LP leads AUROC at 0.753 while VerbConf leads ρ at 0.337) reflects differing sensitivity to binary vs. ordinal discrimination.

V LLaMA-3.2-3B Scale Ablation

To extend scale coverage below 4B, we fine-tune LLaMA-3.2-3B-Instruct on SciERC NER under identical training conditions (LoRA rank=32, α=64, 5 epochs) and evaluate quality estimation and selection (Table 38).

The ρ ranking at 3B (VC > SJ > EM > FK $\approx$ LP) preserves the structural-over-LP ordering from 8B (Table 1); ranking stability holds across two model families spanning 3B–8B. AUROC rankings differ (EM > VC > SJ > FK > LP), consistent with the ρ–AUROC dissociation noted in the main text. Despite 13.63 pp of oracle headroom, no signal achieves significant selection improvement (best: SJ, $+0.61$ pp, p=0.175); VC, the strongest QE signal, yields the worst selection (-1.54 pp), consistent with the correlation-selection gap observed at larger scales.

W Few-NERD Full Results

Table 39 reports the complete Few-NERD INTRA coarse evaluation (Qwen3-8B, LoRA rank=32, 3 epochs, N=8, T=1.0). After gold-filtering (removing instances with no gold entities), the evaluation set contains n=32,565 instances across 8 coarse entity types; 11.9% receive zero F1 under greedy decoding. Instance degeneracy is 13.8% (vs. 17.8% on SciERC, 48.4% on CoNLL).

Key observations. (1) All five signals achieve positive QE (ρ 0.22–0.39); quality estimation works on Few-NERD. VC leads (ρ=0.39), followed by FK (0.31), SJ (0.30), EM (0.29), LP (0.22); the ranking matches SciERC where structural signals also outperform LP for quality estimation. (2) LP is the only signal achieving significant positive selection ($+1.39$ pp, $p<0.001$, CI excludes zero). The sample-level AUROC of 0.578 confirms LP provides per-sample discriminability beyond instance-level quality prediction. (3) Despite positive ρ, all structural signals degrade selection (-1.79 to -3.85 pp), further confirming that the correlation-selection gap is not an artifact of weak QE. (4) Low degeneracy (13.8%) combined with strong LP-quality alignment ($\rho\approx$0.5–0.7) enables LP selection: candidates differ in F1 more often, and LP can rank them effectively. This contrasts with SciERC (17.8% degeneracy but weak alignment $\rho\approx$0.24, LP $+0.61$ pp) and CoNLL (48.4%, LP -0.85 pp).

X Per-Entity-Type LP Selection Analysis

Table 40 stratifies LP selection by entity type on Few-NERD, sorted by degeneracy rate. The degeneracy-gating mechanism holds at entity-type granularity: 7 of 8 types show positive LP selection gains, and the sole negative type (Art, -0.17 pp) has the highest degeneracy (19.8%). The correlation between per-type degeneracy and oracle headroom is strong and significant (Spearman $\rho=-0.881$, p=0.004; exact permutation p=0.005); degeneracy directly constrains the headroom available for selection. Under a conservative df=4 adjustment for within-type dependence, the correlation remains significant (p=0.033). Entity types show near-zero intra-class correlation (ICC $= -0.004$), supporting the independence assumption underlying per-type analysis.

The cross-dataset degeneracy gradient (48.4% $\rightarrow$ 17.8% $\rightarrow$ 13.8%) is inversely associated with LP selection effectiveness at the macro level (n=3

Table 40: Per-entity-type LP selection on Few-NERD INTRA coarse (Qwen3-8B, N=8, T=1.0), sorted by degeneracy. LP Δ: LP selection F1 minus greedy. 7/8 types show positive Δ; Art (highest degeneracy) is the sole exception.

Entity Type	n	Degen.%	Greedy	LP Sel	Δ (pp)	LP p
Event	1,306	6.0	.655	.662	+0.65	.457
Other	3,433	6.6	.637	.685	**+4.73**	.491
Organization	6,647	6.7	.720	.732	+1.21	.500
Building	1,587	7.2	.702	.705	+0.23	.478
Product	2,276	8.2	.656	.677	+2.01	.488
Location	8,023	11.4	.793	.798	+0.53	.520
Person	8,016	13.4	.822	.835	+1.25	.544
Art	1,277	19.8	.756	.754	−0.17	.451

Table 41: CoNLL-2003 per-type LP selection (Qwen3-8B, N=16, T=1.0). All types exceed 20% degeneracy and show negative LP selection, consistent with the dataset's ceiling-regime performance (F1 > 90%).

Entity Type	Degen.%	LP Δ (pp)
MISC	23.2	−2.19
PER	42.6	−1.08
LOC	47.5	−0.62
ORG	54.3	−0.55

Table 42: SciERC per-type LP selection (Qwen3-8B, N=8, T=1.0). Within-dataset, degeneracy alone does not predict LP effectiveness (Spearman ρ=0.14, p=0.79); additional type-specific factors likely modulate selection at this granularity.

Entity Type	Headroom	Degen.%	LP Δ (pp)
Task	16.1 pp	10.1	+2.58
Generic	19.4 pp	10.0	+2.07
Material	13.3 pp	20.0	+0.91
Metric	14.9 pp	12.5	−0.06
Method	15.8 pp	16.3	−0.03
OtherScientificTerm	12.5 pp	6.2	−0.57

Table 43: Calibration baseline results (N=8, T=1.0). Platt and TempScale are monotonic transforms of LP that preserve the within-instance argmax (selection F1 ≡ Raw LP, omitted). MF-Platt uses LP + token count + entity count; p: bootstrap test vs. Raw LP.

		Selection F1		ECE	
Dataset	Greedy	Raw LP	MF-Platt (p)	Raw	Platt
SciERC	.644	.650	.639 (.72)	.260	.048
CoNLL	.909	.898	.901 (.31)	.392	.012
Few-NERD	.771	.785	.778 (.83)	.262	.080

datasets; exploratory).[9] At finer per-type granularity, selection depends on the interaction of degeneracy, oracle headroom, and type-specific LP-quality alignment. A Kruskal-Wallis test finds no significant per-type heterogeneity in LP selection gains on SciERC (p=0.796), consistent with the small per-type sample sizes (n=54–296). Low degeneracy alone does not ensure selection gains: CoNLL MISC (23%, headroom 12.6 pp) and SciERC OtherScientificTerm (6.2%, headroom 12.5 pp) both show negative LP selection despite available headroom; LP correlation strength with the type-specific quality function is an additional prerequisite.

Y Calibration Baseline

Standard calibration methods are monotonic transforms of log-probability and therefore preserve within-instance sample rankings: Platt scaling (Platt, 1999) fits a logistic regression $\sigma(a \cdot \text{LP} + b)$, and temperature scaling (Guo et al., 2017) divides logits by a scalar τ. Since both preserve the argmax, selection F1 is identical to raw LP on all three datasets (Table 43).

To break monotonicity, we fit a multi-feature Platt model (MF-Platt) using LP, token count, and entity count as inputs, trained on a held-out calibration split (20% of the test set, 5-fold cross-validation on the remainder). MF-Platt can reorder candidates within an instance, yet yields no significant selection improvement: Δ=−1.1 pp on SciERC (p=0.72), +0.3 pp on CoNLL (p=0.31), and −0.6 pp on Few-NERD (p=0.83). Platt scaling substantially improves probability calibration (ECE drops from 0.26–0.39 to 0.01–0.08), but better-calibrated probabilities do not translate to better selection. This parallels the supervised signal combination result (§4.5): neither stronger aggregation nor additional features can compensate for weak per-sample discriminative signal.

Z Epoch Confound Analysis

Training convergence modulates degeneracy. Table 44 compares the 3-epoch model (used throughout the main text) with a 5-epoch variant on Few-NERD (Qwen3-8B, N=8, T=1.0). Extended training increases greedy F1 by +2.95 pp (0.748 → 0.777) but nearly triples instance degeneracy (11.1% → 31.3%), reducing the fraction of instances with non-degenerate outputs from 85.6% to 68.7%. LP selection gains drop correspondingly (+1.37 pp → +0.62 pp), consistent with the degeneracy-gating mechanism identified in the main text (§5). Within-instance LP-quality

[9] Partial Spearman correlation controlling for entity-type frequency yields ρ=−0.47 (vs. raw ρ=−0.38); type frequency does not confound the degeneracy–LP relationship.

Table 44: Training convergence confound on Few-NERD (Qwen3-8B, N=8, T=1.0). Evaluated on the exp_027 5K subsample (gold-filtered n=4,351); main-table results (Table 3) use the full test set (n=32,565, gold-filtered), yielding slightly different degeneracy (13.8% vs. 11.1% here). "Degen.": constant-F1 instance degeneracy (gold-filtered). "Valid": fraction of instances with non-degenerate outputs. "ρ mean/med": within-instance LP-quality correlation. The 3-epoch model corresponds to standard early stopping; the 5-epoch model is a controlled ablation.

Metric	3-epoch	5-epoch
Greedy F1	0.748	0.777
Instance degen. (%)	11.1	31.3
Valid instances	3,723 / 4,351 (85.6%)	2,988 / 4,351 (68.7%)
LP Δ (pp)	+1.37	+0.62
Within-inst. ρ (mean)	0.524	0.407
Within-inst. ρ (median)	0.667	0.615

Table 45: SciERC NER epoch curve (Qwen3-8B, N=8, T=1.0). Training duration modulates both instance degeneracy and within-instance LP-quality alignment. "Degen.": constant-F1 instance degeneracy (gold-filtered). "LP Δ": LP selection gain over greedy (gold-filtered; none reaches the ≈1.5 pp MDE). "ρ med": median within-instance LP-quality correlation. Greedy F1 uses span-based evaluation consistent with Tables 1–3; ρ and LP Δ use text-based matching consistent with Table 23.

Metric	3-epoch	5-epoch	10-epoch
Greedy F1	0.650	0.644	0.649
Instance degen. (%)	6.4	17.8	47.8
LP Δ (pp)	−1.24	+0.64	+1.02
Within-inst. ρ (median)	0.319	0.262	0.112
Within-inst. ρ (mean)	0.273	0.179	0.066
% positive ρ	76.4	64.1	54.0

alignment is relatively stable under this convergence shift (mean ρ: $0.524 \rightarrow 0.407$; median ρ: $0.667 \rightarrow 0.615$), consistent with alignment being determined primarily by task structure rather than training duration.

Evaluation loss increases after epoch 3 ($0.032 \rightarrow 0.041$ at epoch 5), and the best checkpoint by validation loss occurs at step 32,000 (approximately epoch 3.9). Standard early stopping practice thus naturally selects the low-degeneracy regime where LP selection is most effective. This finding refines the narrative: degeneracy is not a fixed property of datasets but is modulated by the degree of training convergence. Over-trained models compress their output distributions, collapsing diversity even on datasets that support it under well-regularized training.

SciERC epoch curve. Table 45 extends the analysis to SciERC with a 3-point curve (3, 5, 10 epochs; Qwen3-8B, N=8, T=1.0). Degeneracy increases monotonically (6.4% → 17.8% → 47.8%), and within-instance LP-quality correlation degrades in tandem (median ρ: $0.319 \rightarrow 0.262 \rightarrow 0.112$; mean ρ: $0.273 \rightarrow 0.179 \rightarrow 0.066$), confirming that overtraining erodes both diversity and LP alignment simultaneously. The fraction of instances with positive LP-quality correlation also declines (76.4% at 3 epochs → 64.1% at 5 epochs → 54.0% at 10 epochs). LP selection does not yield significant gains at any training duration (−1.24, +0.64, +1.02 pp; all below the ≈1.5 pp MDE), despite oracle headroom ranging from 8.4 to 15.0 pp. The 3-epoch model has higher within-instance alignment (ρ=0.319) than the 5-epoch model (ρ=0.262), yet LP selection is worse (−1.24 vs. +0.64 pp): at 6.4% degeneracy, more non-degenerate instances enter the candidate pool, but LP ranking precision is too low to exploit them, so the broader pool hurts rather than helps. This illustrates that low degeneracy does not ensure selection gains; LP-quality alignment determines selection direction. At 10 epochs, the model retains competitive greedy F1 (0.649 vs. 0.650 at 3 epochs) yet conditions are unfavorable: 47.8% degenerate instances and ρ=0.112. The gradient confirms that degeneracy is convergence-driven rather than a fixed dataset property: extended training compresses output distributions, degrading both diversity and within-instance alignment. However, the epoch-to-epoch alignment variation ($\Delta\rho_{\text{median}}$=0.057) is small relative to the cross-dataset gap (SciERC ∼0.3 vs. Few-NERD ∼0.5), consistent with alignment being primarily dataset-intrinsic: training convergence modulates alignment at the margin but cannot shift a dataset into the strong-alignment regime. Even at the most favorable epoch (3-epoch, ρ=0.319), SciERC alignment remains below the ∼0.5 threshold associated with successful LP selection on Few-NERD.

Limitations. The Few-NERD epoch analysis covers 2 points (3 vs. 5 epochs); the SciERC curve provides a 3-point gradient showing monotonic degeneracy increase with simultaneous within-instance alignment degradation. CoNLL epoch variation remains unverified; high-degeneracy datasets (CoNLL at 59–89%) may exhibit a ceiling effect where additional training has limited

marginal impact. All Few-NERD results in the main text use the 3-epoch model (standard early stopping); the 5-epoch comparison is a controlled ablation. All SciERC results use the 5-epoch model; the 3- and 10-epoch variants serve as controlled ablations.

Z.1 Degeneracy-Aware LP Selection

The non-tied subset analysis (Table 22) identifies within-instance LP range (max minus min LP across N samples) as a per-instance degeneracy proxy: instances with near-zero LP range carry no discriminative signal. Degeneracy-aware LP selection (DALS) operationalizes this: when LP range exceeds τ=0.05 (the threshold independently motivated by the non-tied analysis; Appendix J), select the highest-LP sample; otherwise fall back to greedy. On SciERC, DALS yields $+0.71 \pm 0.48$ pp (3-seed mean; individual seeds: $+0.19$ to $+1.14$ pp); on Few-NERD, $+0.30$ pp ($p<0.001$; gain remains positive across $\tau \in [0.01, 0.15]$); on CoNLL, where pure LP selection degrades F1 (-0.85 pp), DALS shows comparable degradation (Qwen: -0.87 pp; LLaMA: neutral). DALS improves over pure LP selection when degeneracy is moderate, provides partial damage control when LP selection is harmful, and defaults to greedy when LP range signals no information. While gains are modest and seed-sensitive, LP range serves as an operationally accessible degeneracy proxy, though the mixed cross-dataset results confirm that degeneracy gating alone cannot ensure positive selection outcomes.

LOO τ sensitivity. Leave-one-out (LOO) cross-validation across datasets reveals that the optimal τ spans two orders of magnitude: τ^*=0.001 on Few-NERD, 0.05 on SciERC, and 0.10 on CoNLL. LOO regret (the gap between oracle and LOO-selected τ) ranges from 0.51 to 1.21 pp. The best single cross-dataset τ (τ=0.05) matches the independently motivated non-tied threshold, but yields dataset-specific suboptimality: even with oracle τ, CoNLL remains at -0.31 pp. This sensitivity reinforces the diagnostic framing of DALS: per-dataset τ tuning is required for positive gains, and even optimally tuned DALS cannot overcome weak LP-quality alignment on high-degeneracy datasets.

Z.2 Conditional Decomposition of LP Selection

To test whether LP-quality alignment drives selection gains within datasets, we restrict LP selection to the favorable subset: non-degenerate instances whose within-instance LP range exceeds the dataset-specific median (SciERC: 0.040 nats, CoNLL: 0.062, Few-NERD: 0.067). Table 46 reports results across three datasets.

Table 46: LP selection on the favorable subset (non-degenerate instances with LP range above the dataset-specific median) per dataset (Qwen3-8B, N=8, T=1.0). CI: bootstrap 95% confidence interval.

Dataset	LP Δ (pp)	95% CI	Interpretation
SciERC	$+2.47$	$[+0.30, +4.70]$	significant positive
Few-NERD	$+0.63$	$[-0.02, +1.30]$	marginal positive
CoNLL	-1.78	$[-2.83, -0.76]$	significant negative

On SciERC, restricting to the favorable subset recovers a significant LP gain, consistent with LP-quality alignment driving selection within the non-degenerate pool. On Few-NERD, the trend is directionally consistent but the confidence interval crosses zero; the full-dataset result ($+1.39$ pp, $p<0.001$) reflects the larger sample size rather than a stronger per-instance effect. CoNLL remains negative even on the favorable subset, consistent with the ceiling-regime interpretation: at >90% greedy F1, residual errors are idiosyncratic and LP cannot rank corrections.

Z.3 Entity-Token LP Decomposition

Full-sequence LP averages over all output tokens, diluting entity-relevant signal with formatting and schema tokens. Entity-token LP restricts the average to tokens within entity text and type spans, yielding $4\times$ higher within-instance variance (median 0.000644 vs. 0.000161). Table 47 compares the two aggregation strategies on SciERC (Qwen3-8B, N=8, T=1.0).

Table 47: Full-sequence vs. entity-token LP on SciERC NER (Qwen3-8B, N=8, T=1.0). Entity-token LP has higher variance and stronger correlation, yet selection degrades.

Aggregation	Med. var.	ρ	AUROC	Δ (pp)
Full-sequence	0.000161	0.297	0.642	$+0.35$
Entity text+type	0.000644	0.440	0.719	-0.73

Despite stronger correlation ($+48\%$) and higher AUROC, entity-token LP produces worse selection,

reinforcing that compression operates at the representation level rather than token aggregation: the correlation-selection gap persists even with a more targeted signal.

Z.4 Entity-Level Majority Voting

Entity-level majority voting constructs a merged output by including all (text, type) entity tuples that appear in at least k of N samples (with $k = N/2$ as the default majority threshold). Unlike best-of-N selection (which picks one complete sample), merge combines evidence across samples at entity granularity. We sweep $k \in \{2, 3, \ldots, 8\}$ on each dataset (Qwen3-8B, N=8, T=1.0) and report the best threshold per dataset.

Table 48: Entity-level majority voting results (Qwen3-8B, N=8, T=1.0). "Greedy": T=0 baseline. "Best merge": optimal k per dataset from sweep $k \in \{2, \ldots, 8\}$. "Δ": merge F1 minus greedy. "Recovery": Δ / (oracle − greedy). Only SciERC shows positive recovery; even with oracle k, CoNLL and Few-NERD degrade.

Dataset	Greedy	Best merge (k^*)	Δ (pp)	Recovery (%)
SciERC	0.644	0.660 (k=3)	+1.57	+11.9
CoNLL	0.907	0.903 (k=3)	−0.38	−9.0
Few-NERD	0.748	0.744 (k=2)	−0.43	−3.6

On degenerate subsets (instances where all N samples yield identical F1), merge is uniformly negative across all three datasets, confirming that entity-level aggregation cannot bypass the diversity bottleneck identified in the main text (§5).